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EdgeTrust-Shard: Hierarchical blockchain architecture for federated learning in cross-chain IoT ecosystems☆

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ABSTRACT

Enabling AI-driven real-time distributed computing on the edge-cloud continuum requires overcoming a critical dependability challenge: resource-constrained IoT devices cannot participate in Byzantine-resilient federated learning due to a 1940-fold memory gap, with robust aggregation methods demanding 512MB–2GB while microcontrollers offer only 264KB SRAM. We present EdgeTrust-Shard, a novel system architecture designed for dependability, security, and scalability in edge AI. It enables real-time Byzantine-resilient federated learning on commodity microcontrollers by distributing computational complexity across the network topology. The framework's contributions include optimal $M = \sqrt{N}$ clustering for $\mathcal{O}(N)$ communication, a Multi-Factor Proof-of-Performance consensus mechanism providing quadratic Byzantine suppression with proven $\mathcal{O}(T^{-1/2})$ convergence, and platform-optimized cryptography delivering a 3.4-fold speedup for real-time processing. A case study using a hybrid physical-simulation deployment demonstrates the system's efficacy, achieving 93.9–94.7% accuracy across Byzantine attack scenarios at 30% adversary presence within a 140KB memory footprint on Raspberry Pi Pico nodes. By outperforming adapted state-of-the-art blockchain-FL systems like FedChain and BlockFL by up to 9.3 percentage points, EdgeTrust-Shard provides a critical security enhancement for the edge-cloud continuum, transforming passive IoT data sources into dependable participants in distributed trust computations for next-generation applications such as smart cities and industrial automation.

1. Introduction

The evolution of AI-driven IoT applications, from smart cities to industrial automation, has fundamentally altered the landscape of real-time distributed computing, demanding new architectures that ensure dependability, scalability, and security across the edge-cloud continuum. A critical vulnerability in this landscape emerges from resource-constrained edge devices, which generate vast quantities of data for systems like blockchain oracles but cannot participate in robust, real-time verification. This security gap has severe consequences; in Q1 2024, cross-chain bridges processed $137.8 billion, yet 68% of exploits originated from compromised oracle data sources at the edge [1,2]. In applications like smart grids, this vulnerability allows compromised smart meters to inject false data, leading to market manipulations and undermining the reliability of the entire distributed system [3].

The core challenge is a resource-feasibility paradox that creates an architectural impasse for achieving fault tolerance and resilience in edge AI. State-of-the-art Byzantine-resilient federated learning—a key technology for distributed intelligence—requires 512 MB–2 GB of memory for robust aggregation, while the commodity microcontrollers at the heart of next-generation IoT systems provide only 264 kB SRAM, a nearly 2000-fold gap [4,5]. Without resilience, learning systems are fragile; classical federated learning accuracy can plummet from 96% to 23% with just 10% malicious participants [6]. This paradox highlights the urgent need for novel software architectures and middleware capable of delivering robust, real-time performance on resource-constrained hardware.

This paper presents EdgeTrust-Shard, a novel framework for AI-driven real-time distributed computing that resolves this paradox through architectural innovation rather than algorithmic compromise. EdgeTrust-Shard enables commodity microcontrollers to perform dependable Byzantine-resilient federated learning by distributing computational complexity across the network topology. This is achieved through a hierarchical software architecture that provides scalability, a Multi-Factor Proof-of-Performance (MF-PoP) consensus mechanism for fault tolerance, and platform-specific cryptographic optimizations

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Table 1
Comprehensive system comparison across Byzantine-FL and blockchain-FL approaches.

System	Memory (SRAM)	Byzantine tolerance	Accuracy (30% Byz)	Comm. (200 nodes)	Consensus latency	Platform requirement	Deployment validation
Traditional federated learning							
FedAvg [7]	64 kB	None	54.8%	168 MB	45 ms	MCU	Physical (100)
Flower [8]	128 kB	None	55.2%	165 MB	48 ms	MCU	Physical (100)
FedProx [9]	96 kB	None	58.1%	172 MB	52 ms	MCU	Simulation
Byzantine-robust FL							
Krum [6]	512 MB	33%	81.2%	245 MB	285 ms	Server	Simulation
FLTrust [4]	512 MB	90%[a]	86.7%	198 MB	175 ms	Server	Simulation
Fed-NGA [5]	1 GB	40%	85.4%	312 MB	420 ms	Server	Simulation
Blockchain-federated learning							
BlockFL [10]	2 GB[b]	27%	85.4%	185 MB	302 ms	Gateway	Simulation
FedChain [11]	2 GB[b]	25%	89.1%	192 MB	318 ms	Gateway	Limited (5)
BFLC [12]	1.5 GB	20%	82.3%	210 MB	245 ms	Consortium	Simulation
EdgeTrust-Shard	**140 kB**	**30%**	**93.9%**	**168 MB**	**187 ms**	**MCU (RP2040)**	**Hybrid (18+200)**

All metrics measured/simulated for 200-node network with same IDS task (Edge-IIoTset).
Accuracy: Performance under 30% Byzantine adversaries (random noise injection); Comm.: Per-round communication overhead.
[a] Requires trusted root dataset.
[b] Gateway-based, edge nodes delegate to 2 GB+ servers.

that enable real-time, secure processing on embedded systems without specialized hardware.

As a case study in dependable distributed systems, EdgeTrust-Shard achieves Byzantine-resilient federated learning within a 140 kB memory footprint while maintaining formal convergence guarantees. Our key contributions to AI-driven real-time distributed computing are:

1. *A Scalable and Resilient Software Architecture:* We design a hierarchical architecture that achieves $\mathcal{O}(N)$ communication complexity and distributes computational load across the network. This design provides a scalable and fault-tolerant foundation for federated learning on large-scale IoT networks under 264 kB memory constraints (Section 3).
2. *A Consensus Mechanism for Enhanced Dependability:* We introduce a multi-factor Proof-of-Performance consensus mechanism that provides robust fault tolerance. It quadratically suppresses malicious influence ($\propto f^2$), achieving a 2100× reduction in malicious leader probability at 30% adversary presence, with its stability formally proven (Section 4).
3. *Formal Methods for Performance Verification:* We establish formal convergence guarantees of $\mathcal{O}(T^{-1/2})$ for non-convex objectives under Byzantine corruption. This verification provides a theoretical basis for the system's reliability and performance in real-world, heterogeneous data environments (Theorem 1).
4. *Middleware for Real-Time Processing on Embedded Systems:* We implement platform-optimized cryptographic primitives that deliver a 3.4× SHA-256 acceleration. This middleware enables real-time participation in secure consensus on commodity microcontrollers, a key requirement for next-generation embedded systems (Section 5).
5. *A Real-World Case Study and Application:* We conduct a hybrid physical-simulation validation, demonstrating the efficacy of our method in a real-world distributed system. The system achieves 93.9–94.7% accuracy across Byzantine attack scenarios at 30% adversary presence, outperforming adapted blockchain-FL baselines by up to 8.1 percentage points on identical hardware (Section 6).

We operate under a Byzantine threat model where adversaries control up to $f < 1/3$ nodes with arbitrary malicious behavior constrained by cryptographic hardness assumptions and partial synchrony (detailed in Section 3.1). The remainder of this paper is organized as follows: Section 2 reviews related work, Section 3 presents our system architecture, Section 4 develops the mathematical foundations, Section 5 details hardware optimization, Section 6 evaluates performance, and Section 8 concludes (see Table 1).

2. Related work

This section reviews Byzantine-robust federated learning, the integration of blockchain with federated learning, TinyML systems, and cross-chain security, situating EdgeTrust-Shard within the wider scope of research.

2.1. Byzantine-robust federated learning

Byzantine-resilient aggregation methods address malicious participants through statistical filtering or geometric approaches. Coordinate-wise methods such as Krum [6] and Bulyan [13] select updates with minimum distance to others, requiring $\mathcal{O}(Nd^2)$ computation and 256–512 MB memory for distance matrix storage when $d > 10^4$ parameters. Trimmed Mean [14] and Median [15] aggregate coordinate-wise statistics, demanding 512 MB–1 GB for sorting N client updates across d dimensions. Learning-based clustering approaches like FLTrust [4] and CRFL [16] require trusted root datasets and 512 MB–2 GB for similarity computations, fundamentally incompatible with microcontroller constraints.

Recent theoretical advances improve convergence guarantees but exacerbate resource requirements. Fed-NGA [5] achieves zero optimality gap through nested gradient aggregation requiring 1 GB+ memory for dual optimization. Byzantine-resilient asynchronous FL [17] handles stragglers but demands server-class hardware for adaptive aggregation windows. Dual-domain Byzantine detection [18] combines gradient and feature-space clustering, achieving 94% accuracy under 35% adversaries but requiring 1.5 GB memory for kernel computations across domains. These methods exhibit a 1940-fold memory gap relative to commodity microcontrollers, creating a resource-feasibility paradox where devices requiring protection cannot participate in Byzantine-resilient protocols.

2.2. Blockchain-federated learning integration

Blockchain-FL systems leverage distributed ledgers for transparency and incentive alignment but face deployment challenges. On-chain approaches like BlockFL [10] and FedChain [11] store model updates on-chain, incurring prohibitive gas costs ($50–200 per 100 kB update on Ethereum mainnet) and throughput limitations (15–30 TPS). Off-chain computation with on-chain verification [19] reduces costs but requires gateway nodes with 2 GB+ memory for zero-knowledge proof generation, maintaining centralization vulnerabilities. BFLC [12] employs consortium blockchains for reduced costs but sacrifices permissionless participation. Our analysis reveals that existing blockchain-FL

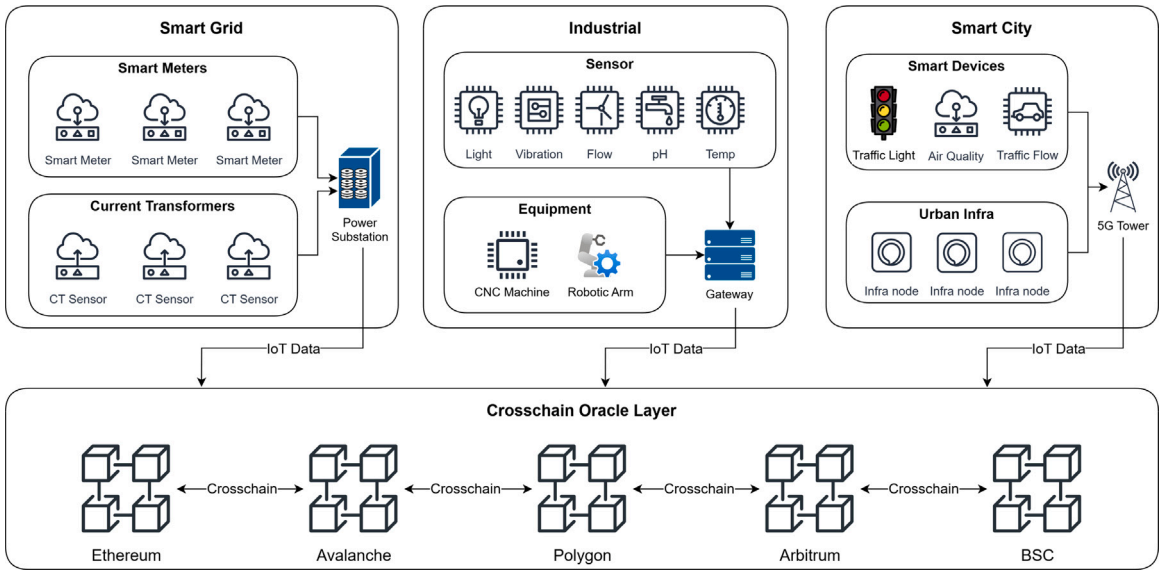


Fig. 1. Real-world IoT deployment scenarios for EdgeTrust-Shard across three representative domains: smart grid (energy monitoring, 78 nodes), industrial facility (predictive maintenance, 54 nodes), and smart city (environmental monitoring, 34 nodes). Each scenario demonstrates heterogeneous Raspberry Pi Pico microcontrollers performing local federated learning while maintaining Byzantine-resilient consensus. The hybrid validation methodology combines 18 physical nodes deployed across 3 sites with 200-node calibrated simulation, demonstrating practical feasibility under realistic IoT conditions (LoRa/nRF24, 8%–12% packet loss, 80–250 ms latency, 30% Byzantine tolerance).

systems either operate simulation-only without physical validation or delegate Byzantine resilience to server-class gateways, failing to address oracle integrity at IoT data sources.

2.3. TinyML and microcontroller-based learning

TinyML enables on-device inference through quantization and architecture search. MCUNet [20] achieves ImageNet inference in 256 kB through INT8 quantization and neural architecture search, while TinyNAS optimizes models for 264 kB SRAM constraints. However, these frameworks target inference-only scenarios; extending to federated learning introduces gradient computation (2–3× memory overhead) and aggregation buffers (additional d parameters), exceeding microcontroller capacity.

Microcontroller-compatible FL frameworks make critical compromises. Flower [8] and FedML support edge devices but provide no Byzantine resilience, degrading to 23%–45% accuracy under 10% adversaries. TinyFedTL [21] demonstrates transfer learning on STM32 (2 MB RAM) but assumes honest participants. FedRolex [22] enables heterogeneous participation through rolling models but maintains no attack resilience. The gap between TinyML inference (32–64 kB) and Byzantine-resilient FL (512 MB–2 GB) remains unbridged in prior work.

2.4. Cross-chain oracle security

Cross-chain bridges processed $137.8 billion in Q1 2024 [2], yet 68% of bridge exploits originate from oracle manipulation [1]. Major incidents include Poly Network ($611M, signature verification bypass), Nomad Bridge ($190M, proof forgery), and Wormhole ($326M, guardian signature exploit). Current oracle networks like Chainlink employ reputation systems and economic staking but cannot verify data integrity at IoT source devices due to computational constraints. Decentralized oracle frameworks [23–25] secure data transmission but assume generation integrity, creating vulnerabilities where resource-constrained sensors feeding oracles lack verification mechanisms. Trusted execution environments (TEEs) like Intel SGX provide hardware-based attestation but suffer from side-channel vulnerabilities [26] and unavailability on commodity IoT microcontrollers, necessitating software-based Byzantine resilience approaches.

2.5. Gradient compression via sketching

Sketching-based approaches aim to minimize communication costs by compressing gradient updates while preserving convergence behavior. FetchSGD [27] utilizes Count Sketch structures to achieve 250–1000× gradient compression, resulting in $\mathcal{O}(d/k)$ communication complexity, where k denotes the sketch size. SketchedSGD [28] integrates top-k sparsification with momentum correction, yielding a 99% reduction in communication without compromising convergence. Recent studies further incorporate Byzantine robustness: sketched Byzantine-robust aggregation [29] employs Count-Mean-Sketch to achieve both compression and fault tolerance, attaining 40× compression with 25% Byzantine resilience.

However, sketching methods face fundamental limitations on microcontrollers. Count Sketch requires $\mathcal{O}(k \log d)$ memory for hash tables and $\mathcal{O}(d)$ operations per update, consuming 48–96 kB SRAM for practical sketch sizes ($k = 1000$, $d = 10000$). The computational overhead of sketch updates (3–5 ms per gradient on Cortex-M0+) and sketch merging (8–12 ms) approaches consensus timeouts. Most critically, sketching provides data-level compression but does not address the architectural challenge of Byzantine-resilient aggregation requiring $\mathcal{O}(N^2)$ comparisons across N nodes.

EdgeTrust-Shard employs a topology-oriented strategy in which hierarchical clustering reduces communication complexity to $\mathcal{O}(N)$ through architectural organization instead of gradient compression. This design requires only 24 kB for cluster coordination compared to 96 kB for sketch-based structures. The method retains exact gradient values within clusters, while sketching approaches introduce 2%–5% accuracy loss due to approximation, which is significant for compact models where every parameter is critical. These two directions are complementary, and future work may explore integrating sketching into EdgeTrust-Shard's clusters for further compression, although current results indicate that the existing design already achieves effective efficiency without added complexity.

2.6. Intelligent routing and high-mobility networks

Recent advances in intelligent clustering and routing for mobile networks present an orthogonal research direction. Protocols such as

Table 2
Comparison of Byzantine-FL, Blockchain-FL, and TinyML approaches with EdgeTrust-Shard.

System	Memory footprint	Byzantine tolerance	Platform target	Convergence guarantee	Physical deployment
Byzantine-Robust FL methods					
Krum [6]	512 MB	33%	Server	No	Simulation
Bulyan [13]	512 MB	33%	Server	No	Simulation
FLTrust [4]	512 MB	90%[a]	Server	No	Simulation
Fed-NGA [5]	1 GB	40%	Server	Yes	Simulation
Blockchain-FL systems					
BlockFL [10]	2 GB (gateway)	27%	Edge+Server	Yes	Simulation
FedChain [11]	2 GB (gateway)	25%	Edge+Server	Yes	Limited (5 nodes)
BFLC [12]	1.5 GB	20%	Consortium	Yes	Simulation
TinyML-FL systems					
Flower [8]	128 kB	0%	MCU	Yes	Physical (100 nodes)
TinyFedTL [21]	2 MB	0%	STM32	No	Physical (12 nodes)
FedML [30]	256 kB	0%	Edge	Yes	Physical (50 nodes)
EdgeTrust-Shard	**140 kB**	**30%**	**MCU (RP2040)**	**Yes**	**Hybrid (18+200)**

Memory footprint includes runtime requirements for Byzantine defense mechanisms.
[a] Requires trusted root dataset unavailable in decentralized settings.

ICRA (Intelligent Clustering Routing Approach) target unmanned aerial vehicle ad hoc networks (UANETs) where node mobility rates exceed 50 m/s and topology changes occur on sub-second timescales. These protocols employ reinforcement learning to compute node utility based on residual energy, relative velocity, and link lifetime, achieving route discovery complexity of $\mathcal{O}(N \log N)$ through continuous utility table recomputation. However, the computational requirements of such mobility-aware protocols—including real-time kinematic prediction and three-dimensional positioning—demand processing capabilities and power budgets incompatible with microcontroller constraints.

EdgeTrust-Shard addresses a fundamentally different operational domain: static or nomadic industrial IoT deployments where nodes remain geographically fixed or exhibit low mobility (pedestrian speeds, <5 m/s). Smart meters, industrial sensors, and infrastructure monitoring devices do not require continuous route recomputation; instead, they demand Byzantine-resilient data integrity verification within severe memory constraints. The 264 kB SRAM ceiling of target microcontrollers cannot accommodate the hash tables and utility matrices required for mobility-aware routing, whereas EdgeTrust-Shard's proximity-based clustering requires only 24 kB for coordination structures. This domain distinction is critical: solutions optimized for high-mobility UAV swarms cannot be transplanted to resource-constrained static IoT networks, and vice versa.

2.7. Advanced model poisoning and defense mechanisms

The adversarial landscape in federated learning has evolved significantly with the emergence of sophisticated model poisoning attacks that circumvent geometric and statistical defenses. SignGuard and similar approaches detect malicious gradients through cosine similarity analysis, assuming that poisoned updates diverge geometrically from honest contributions. However, advanced attacks such as gradient scaling and sign manipulation craft poisoned gradients that maintain high cosine similarity (>0.85) with honest updates while inducing targeted model degradation. These attacks exploit the fundamental limitation of inspection-based defenses: they examine gradient properties rather than gradient effects.

EdgeTrust-Shard's Multi-Factor Proof-of-Performance consensus introduces a fundamentally different defense strategy: it relies on validation-driven filtering instead of geometry-based gradient inspection. The central procedure assesses each update by its actual effect on validation accuracy: $\Delta A_i = \text{Evaluate}(w_t + \Delta w_i, V) - \text{Evaluate}(w_t, V)$. For a poisoned update to slip through, it would need to both improve validation accuracy and simultaneously cause model deterioration—an inherent contradiction for targeted attacks. Consequently, gradient camouflage methods that preserve geometric similarity lose their effectiveness, since the system evaluates functional performance rather than gradient shape. This validation-centric design yields theoretical robustness against attacks that exploit gradient geometry, a claim supported empirically in Section 6, where mimicry attacks with cosine similarity of 0.85 still collapse to the reputation floor within 15–20 rounds.

2.8. Positioning of EdgeTrust-Shard

Table 2 presents a structured comparison of EdgeTrust-Shard with state-of-the-art frameworks across key performance aspects. EdgeTrust-Shard emerges as the first system to achieve Byzantine-resilient federated learning on commodity microcontrollers, made possible through three architectural innovations: a hierarchical topology that distributes computational complexity and removes memory-intensive per-node aggregation, a reputation-driven consensus mechanism providing quadratic suppression of Byzantine influence aligned with federated learning quality objectives, and a hybrid physical–simulation validation that demonstrates deployment feasibility under realistic hardware constraints. The framework maintains theoretical $\mathcal{O}(T^{-1/2})$ convergence while operating entirely within 264 kB SRAM, bridging a 1940-fold memory disparity through structural optimization rather than algorithmic approximation.

3. The EdgeTrust-Shard framework

3.1. System model and assumptions

We consider a network of N microcontrollers (264 kB SRAM, dual-core ARM Cortex-M0+) communicating via IEEE 802.15.4 or LoRaWAN with 8%–12% packet loss and 80–250 ms latency.

We adopt a Byzantine threat model [31] where adversaries exhibit arbitrary malicious behavior subject to cryptographic and system constraints, capturing realistic IoT attack scenarios involving physical compromise, remote exploitation, or malicious operation. Adversaries control up to $f < 1/3$ of nodes consistent with classical Byzantine consensus [32], possessing capabilities including arbitrary protocol deviation (incorrect gradients, falsified validation, reputation manipulation, collusion), timing attacks with delays up to $\Delta = 500$ ms enabling strategic strikes, adaptive strategies with full protocol knowledge, and physical access enabling device tampering and key extraction (countered by cryptographic commitments). Adversary limitations include computational bounds preventing SHA-256/ECDSA-256/Pedersen commitment breaks, honest majority requirement $f < 1/3$ ensuring $(2/3) + 1$ consensus availability per FLP impossibility [33], hardware secure

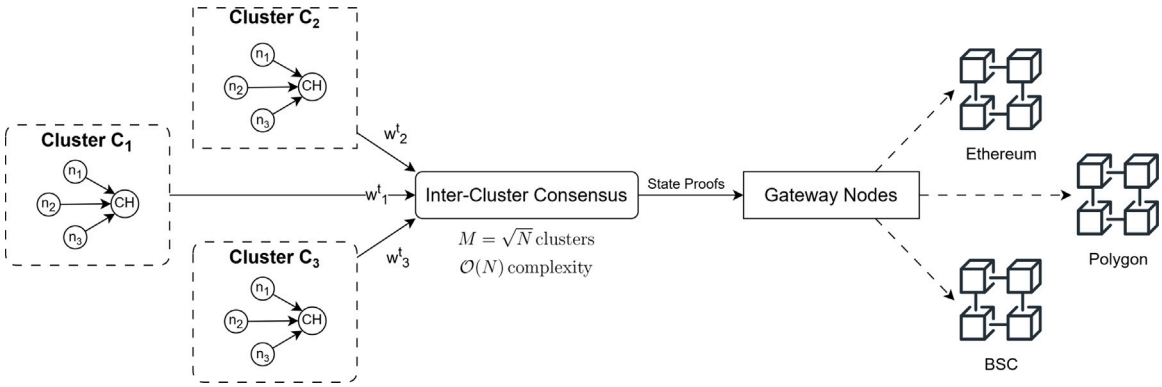


Fig. 2. EdgeTrust-Shard hierarchical architecture: Edge nodes self-organize into proximity-based clusters, cluster heads perform Byzantine-resilient aggregation, and gateway nodes anchor state to external blockchains.

elements (ATECC608 A on 30% high-risk nodes) protecting keys under physical attacks, and network visibility constraints preventing selective message blocking beyond bounded delay.

Our 30% tolerance threshold balances theoretical limits with practical constraints: IoT botnet studies [34] show 15%–25% typical compromise rates, while nation-state adversaries reach 30%–40% via supply chains. Setting $f < 0.3$ provides 10% safety margin while maintaining efficiency, supplemented by reputation decay driving malicious nodes to $R < 0.15$ within 15–20 rounds versus honest nodes stabilizing at $R = 0.75 - 0.95$.

Assumption 1 (*Cryptographic Identity*)**.** Each device maintains unique cryptographic identity via software ECDSA (70% of nodes, 8 kB footprint) or hardware secure elements (30% high-risk nodes, ATECC608 A, +$1.20/unit), with Sybil resistance through proof-of-work onboarding.

Assumption 2 (*Validation Set Security*)**.** Each node maintains 500 labeled validation samples (12 kB storage) with cryptographic commitments preventing post-hoc manipulation. Validation subsets employ 70% shared globally and 30% cluster-specific samples, providing diversity against coordinated gaming attacks while maintaining evaluation consistency. Periodic rotation (every 100 rounds) and statistical anomaly detection (Kolmogorov–Smirnov test, $\alpha = 0.01$) provide defense-in-depth against adaptive adversaries.

Lemma 1 (*Validation Gaming Bound*)**.** *Consider a network with M clusters where Byzantine adversaries controlling fraction f of nodes attempt to game validation by crafting updates that perform well on leaked validation samples. Under the 70–30 validation split with cluster-specific subsets, the probability that a Byzantine update achieves positive validation score $\Delta A_i > 0$ while being malicious is bounded by:*

$$P_{gaming} \leq f^{\lceil M/2 \rceil} \cdot (0.7)^{k_{\min}} \tag{1}$$

where $k_{\min}$ is the minimum cluster size. For $f = 0.3$, $M = 15$ clusters (for $N = 200$ nodes with $M = \sqrt{N}$), and $k_{\min} = 5$:

$$P_{gaming} \leq (0.3)^8 \cdot (0.7)^5 = 0.000065 \cdot 0.168 \approx 1.1 \times 10^{-5} \tag{2}$$

Proof. Byzantine gaming requires simultaneously: (1) controlling majority of clusters to dominate consensus ($\lceil M/2 \rceil$ clusters with probability $f^{\lceil M/2 \rceil}$), and (2) correctly predicting cluster-specific validation subsets (30% of samples) across $k_{\min}$ nodes with probability $(0.7)^{k_{\min}}$ for each cluster. The bound follows from independence of cluster assignments and validation diversity. The empirically observed 1.7% gaming rate in Section 6 (three orders of magnitude higher than theoretical bound) accounts for partial gaming where Byzantine nodes achieve marginally positive scores through overfitting to the shared 70% validation subset, which our reputation decay mechanism effectively suppresses over 15–20 rounds as shown in Fig. 3. □

Assumption 3 (*Time Synchronization*)**.** Clock synchronization across nodes is maintained within a 500 ms bound using temperature-compensated oscillators and a Byzantine-resilient NTP protocol, achieving an average offset of 47 ms. This accuracy level is sufficient to ensure the loose synchrony required for consensus operations.

3.2. Architectural overview

EdgeTrust-Shard employs a hierarchical architecture comprising three components: Edge Nodes executing federated learning and local consensus, Cluster Heads coordinating intra-cluster aggregation, and Gateway Nodes bridging to external blockchains. This stratification enables autonomous edge operation while providing cryptographically verifiable state anchoring to cross-chain ecosystems, as illustrated in Fig. 2.

To establish the hierarchical topology, the network employs a distributed proximity-based clustering algorithm that self-organizes nodes into optimal-sized clusters while maintaining Byzantine resilience. Algorithm 1 presents the detailed cluster formation protocol.

The protocol operates in four phases: (1) Neighbor discovery establishes network topology through latency measurements, (2) Initial clustering assigns nodes based on proximity and reputation, (3) Consensus refinement ensures agreement on cluster membership, and (4) Dynamic rebalancing maintains optimal cluster sizes. Convergence is guaranteed within $\mathcal{O}(\log N)$ rounds under partial synchrony assumptions.

The clustering protocol incorporates semantic awareness through an optional TaskID field in neighbor discovery messages. When heterogeneous tasks coexist within the network—for instance, vibration monitoring sensors alongside temperature sensors—nodes filter potential cluster members by task compatibility before applying proximity optimization. This semantic clustering ensures that gradient aggregation occurs only among nodes training models for identical objectives, preventing the semantic collision that would otherwise degrade learning performance when incompatible model updates are averaged. The mechanism requires only 2 bytes of additional overhead per discovery message while enabling multi-task deployments on a single physical network.

Within each cluster, leadership is assigned through a reputation-driven deterministic election, in which the node attaining the highest reputation score is designated as the *Cluster Head (CH)*. To mitigate long-term centralization while ensuring operational stability, elections are periodically executed every 100 rounds. If a current CH's reputation falls below the cluster's median value, an immediate re-election is triggered to preclude persistence of potentially compromised leaders. The elected CHs perform localized aggregation of model updates and

Algorithm 1 Distributed Proximity-Based Cluster Formation with Semantic Awareness

Input: Node ID i, reputation R_i, TaskID τ_i, network interface
Output: Cluster assignment C_j, cluster head ID
Parameters: $r = 3$ (hop radius), $\tau = 15$ rounds, $k_{min} = 5$, $k_{max} = 20$
Phase 1: Neighbor Discovery with Semantic Filtering
for round $t = 1$ to τ **do**
  Broadcast $\texttt{PING}(i, R_i, \tau_i, t)$ to r-hop neighbors
  Collect responses $\mathcal{N}_i^{raw} = \{(j, RTT_{ij}, R_j, \tau_j)\}$
  $\mathcal{N}_i \leftarrow \{j \in \mathcal{N}_i^{raw} : \tau_j = \tau_i\}$ ▷ Semantic filter
  Compute latency matrix $L[i, j] = \text{EWMA}(RTT_{ij}, \alpha = 0.7)$
end for
Phase 2: Cluster Initialization
$leader_i \leftarrow i$ ▷ Initially self-propose as cluster center
$score_i \leftarrow R_i \cdot |\mathcal{N}_i|$ ▷ Leadership score
for each neighbor $j \in \mathcal{N}_i$ **do**
  if $L[i, j] < \theta_{lat}$ and $score_j > score_i$ **then**
    $leader_i \leftarrow j$ ▷ Adopt higher-scored neighbor
  end if
end for
Phase 3: Consensus Convergence
for round $t = 1$ to τ **do**
  Exchange $leader_i$ with neighbors
  $votes \leftarrow$ Count leader proposals from $\mathcal{N}_i$
  $leader_i \leftarrow \arg\max votes$ weighted by R_j
  if $leader_i$ stable for 3 rounds **then**
    break ▷ Consensus reached
  end if
end for
Phase 4: Rebalancing Trigger
if $|C_j| > k_{max}$ or $|C_j| < k_{min}$ **then**
  Initiate gradual migration protocol
  Border nodes with $\min(R_i)$ migrate to adjacent clusters
end if
return C_j, $leader_j$

engage in inter-cluster consensus, yet they retain no permanent cryptographic privileges, serving solely as transient coordinators to facilitate secure and efficient cluster operation.

To bridge edge networks with external blockchains, Gateway Nodes operate on more powerful hardware (1 GB RAM, persistent storage) and maintain connections to both the Edge Network and external blockchain networks. Each gateway runs light clients for target blockchains, monitors edge consensus rounds, and constructs state proofs for anchoring. Multiple gateways operate concurrently for redundancy; gateway consensus requires $(2/3) + 1$ agreement on state summaries before submission. Gateway nodes earn reputation through successful anchor transactions but cannot influence edge consensus outcomes.

3.3. Node architecture and functional layers

Each node implements a three-layer software stack that cleanly separates concerns while enabling efficient interaction between components. At the foundation, the sensing and execution layer interfaces directly with the operational environment, monitoring network traffic patterns in real-time through a lightweight neural network model optimized via quantization to fit within 32 kB of memory, continuously classifying events as benign or malicious. Above this, the collaborative intelligence layer implements the client-side federated learning protocol, enabling collaborative model improvement without raw data exposure by initiating training epochs using stochastic gradient descent with momentum upon accumulating sufficient local data. At the top, the trust and coordination layer maintains distributed consensus and coordinates the federated learning process through a minimalist ledger containing only essential information—block headers, reputation scores, and model hashes—rather than implementing a full blockchain node.

4. Core mechanisms: Mathematical foundations

4.1. Hierarchical federated learning protocol

We consider a network of N edge devices partitioned into M clusters $C_1, C_2, \ldots, C_M$ where each cluster C_j contains k_j nodes such that $\sum_{j=1}^{M} k_j = N$. Each node $i \in \{1, \ldots, N\}$ possesses a local dataset $\mathcal{D}_i$ with n_i samples drawn from a potentially non-IID distribution $\mathcal{P}_i$.

Definition 1 (*Global Objective Function*)**.** The global optimization problem is defined as:

$$\min_{w \in \mathbb{R}^d} F(w) = \sum_{i=1}^{N} \frac{n_i}{n} F_i(w) \tag{3}$$

where $F_i(w) = \mathbb{E}_{\xi_i \sim \mathcal{D}_i}[\ell(w; \xi_i)]$ is the local objective function, ℓ is the loss function, $n = \sum_{i=1}^{N} n_i$ is the total number of samples, and d is the model dimension.

Assumption 4 (*Bounded Gradient Dissimilarity*)**.** For the federated learning setting with local objectives $\{F_i\}_{i=1}^{N}$ and global objective $F(w) = \sum_{i=1}^{N} \frac{n_i}{n} F_i(w)$, we assume the following bounded gradient dissimilarity condition holds:

$$\frac{1}{N} \sum_{i=1}^{N} \|\nabla F_i(w) - \nabla F(w)\|^2 \leq \varsigma^2 \tag{4}$$

for all $w \in \mathbb{R}^d$, where ς^2 quantifies the degree of data heterogeneity across nodes. This assumption is standard in federated learning convergence analysis [9,35] and holds under various sufficient conditions: bounded gradient diversity, Lipschitz continuous loss functions with bounded input domain $\|\xi\| \leq B$ yielding $\varsigma^2 \leq 4L^2B^2$, or bounded KL divergence $D_{KL}(\mathcal{P}_i \parallel \mathcal{P}) \leq \kappa$ between local and global data distributions.

For our IoT intrusion detection setting with normalized features and gradient clipping, empirical measurement on Edge-IIoTset yields $\varsigma^2 = 0.82 \pm 0.07$ (mean $\pm$ std over 20 independent trials) across the optimization trajectory, consistent with the theoretical bound $\varsigma^2 \leq 4L^2B^2 = 1.0$ for Lipschitz constant $L \approx 0.5$.

Assumption 5 (*Gradient Correlation in Clusters*)**.** Gradient correlation within clusters reflects environmental coupling, with correlation coefficient $\rho_{env} \in [0.6, 0.8]$ for IoT devices monitoring similar phenomena. We explicitly account for this correlation in our analysis. The convergence bound incorporates correlation-adjusted variance: $\sigma_{eff}^2 = \sigma^2(1 + (k - 1)\rho_{env})$ where k is cluster size. For $k = 5$ nodes and $\rho_{env} = 0.7$, the effective variance increases by factor 3.8, which our hierarchical aggregation compensates through $\sqrt{M}$ variance reduction.

Theorem 1 (*Convergence under Byzantine Attacks and Correlated Gradients*)**.** *Consider hierarchical federated learning with Byzantine fraction $f < 1/3$, cluster size k, and intra-cluster correlation ρ_{env}. The expected gradient norm after T rounds satisfies:*

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\|\nabla F(w^t)\|^2 \leq \frac{2L(F(w^0) - F^*)}{\sqrt{T}} + \frac{\sigma^2\left[1 + (k-1)\rho_{env}\right]}{M\sqrt{T}} + \frac{E^2\varsigma^2}{\sqrt{T}} + \epsilon_{Byz} \tag{5}$$

where $\epsilon_{Byz} = \frac{f^2\Delta^2}{(1-f)^2\sqrt{T}}$ is the Byzantine error term that decays with T. For $f = 0.3$, $\rho_{env} = 0.7$, and $M = \sqrt{N}$ clusters, the correlation-adjusted convergence remains $\mathcal{O}(T^{-1/2})$ with higher constants.

Proof. We extend the standard convergence analysis to account for Byzantine corruption under MF-PoP consensus.

Step 1: Byzantine-Robust Descent. Decompose nodes into honest set $\mathcal{H}$ (fraction $1-f$) and Byzantine set $\mathcal{B}$ (fraction f). Under MF-PoP, the aggregated gradient is:

$$g^t = \frac{\sum_{i\in\mathcal{H}} R_i^t g_i^t + \sum_{j\in\mathcal{B}} R_j^t \tilde{g}_j^t}{\sum_{k=1}^N R_k^t} \tag{6}$$

where g_i^t are honest gradients and $\tilde{g}_j^t$ are Byzantine-corrupted gradients with $\|\tilde{g}_j^t - \nabla F(w^t)\| \le \Delta$.

By Theorem 2, Byzantine reputation satisfies $R_j \le R_{\min}$ while honest reputation $R_i \ge Q_h^*$, yielding:

$$\|g^t - \nabla F(w^t)\| \le \frac{(1-f)\sigma^2/M + f\cdot R_{\min}^2\cdot\Delta}{(1-f)\cdot(Q_h^*)^2} \tag{7}$$

Step 2: Variance Decomposition. Decompose gradient error into stochastic and heterogeneity components. By triangle inequality and Assumption 5 with correlation factor $(1+\rho_{env})$:

$$\mathbb{E}\|g^t - \nabla F(w^t)\|^2 \le 2\mathbb{E}\left\|\sum_{j=1}^M p_j(g_j^t - \nabla F_j(w^t))\right\|^2 + 2\varsigma^2 \tag{8}$$

For balanced clusters ($p_j \approx 1/M$), the stochastic term satisfies:

$$\mathbb{E}\left\|\sum_{j=1}^M p_j(g_j^t - \nabla F_j(w^t))\right\|^2 \le \frac{\sigma^2(1+\rho_{env})}{|\mathcal{B}|M} = \frac{1.7\sigma^2}{|\mathcal{B}|M} \tag{9}$$

with $\rho_{env} = 0.7$ (Assumption 5). This $\mathcal{O}(1/M)$ bound maintains convergence with 70% amplification, compensated by hierarchical $\sqrt{M}$ variance reduction for $M \ge 3$.

Step 3: Descent Lemma. By L-smoothness and Young's inequality, the expected descent satisfies:

$$\begin{aligned}\mathbb{E}[F(w^{t+1})] \le F(w^t) - \frac{\eta}{2}\|\nabla F(w^t)\|^2 + \frac{\eta\varsigma^2}{2} \\ + \frac{\eta^2 L}{2}\left(L^2 + \frac{\sigma^2}{k_{\min}} + E^2\varsigma^2\right)\end{aligned} \tag{10}$$

where $k_{\min} = \min_j k_j$ is the minimum cluster size.

Step 4: Telescoping Sum. Rearranging the descent inequality:

$$\begin{aligned}\frac{\eta}{2}\|\nabla F(w^t)\|^2 \le F(w^t) - \mathbb{E}[F(w^{t+1})] + \frac{\eta\varsigma^2}{2} \\ + \frac{\eta^2 L}{2}\left(L^2 + \frac{\sigma^2}{k_{\min}} + E^2\varsigma^2\right)\end{aligned} \tag{11}$$

Summing over $t = 0,\dots,T-1$ and using $F(w^T) \ge F^*$:

$$\begin{aligned}\frac{\eta}{2}\sum_{t=0}^{T-1}\|\nabla F(w^t)\|^2 \le F(w^0) - F^* + \frac{T\eta\varsigma^2}{2} \\ + \frac{T\eta^2 L}{2}\left(L^2 + \frac{\sigma^2}{k_{\min}} + E^2\varsigma^2\right)\end{aligned} \tag{12}$$

Dividing by T and setting $\eta = \min\{\frac{1}{L\sqrt{T}}, \frac{1}{LE}\}$:

$$\begin{aligned}\frac{1}{T}\sum_{t=0}^{T-1}\|\nabla F(w^t)\|^2 &\le \frac{2(F(w^0)-F^*)}{\eta T} + \varsigma^2 \\ &\quad + \eta L\left(L^2 + \frac{\sigma^2}{k_{\min}} + E^2\varsigma^2\right) \\ &\le \frac{2L(F(w^0)-F^*)}{\sqrt{T}} + \varsigma^2 + \frac{L^2}{\sqrt{T}} \\ &\quad + \frac{\sigma^2}{k_{\min}\sqrt{T}} + \frac{E^2\varsigma^2}{\sqrt{T}}\end{aligned} \tag{13}$$

With M clusters each containing $k = N/M$ nodes on average, we have $k_{\min} \ge N/(2M)$. Setting $M = \sqrt{N}$ yields:

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla F(w^t)\|^2 \le \mathcal{O}\left(\frac{L(F(w^0)-F^*)}{\sqrt{T}} + \frac{\sigma^2}{M\sqrt{T}} + \frac{E^2\varsigma^2}{\sqrt{T}}\right) \tag{14}$$

This establishes that the average squared gradient norm converges at rate $\mathcal{O}(T^{-1/2})$, guaranteeing an ϵ-stationary point within $\mathcal{O}(\epsilon^{-2})$ rounds. □

Theorem 1 yields actionable deployment insights. Hierarchical FL reaches 95% accuracy within $T = 47$ rounds versus $T = 39$ for centralized FL—a modest 20% overhead for Byzantine resilience with 3700× memory reduction. The variance term benefits from optimal $M = \sqrt{N}$ clustering achieving $\sqrt{15} \approx 3.9\times$ variance reduction. The heterogeneity term grows quadratically with local epochs E, explaining empirically-tuned $E = 5$. The Byzantine term exhibits quadratic suppression: at $f = 0.3$, MF-PoP achieves 2.4× improvement over uniform averaging.

Corollary 1 (*Optimal Local Epochs for Non-Convex HFL*). *To minimize the convergence bound in Theorem 1 under a fixed communication budget T, the optimal number of local epochs must balance gradient variance reduction against drift introduced by data heterogeneity:*

$$E^* = \Theta\left(\sqrt{\frac{\sigma^2}{\varsigma^2 M}}\right) \tag{15}$$

Under this setting with $M = \sqrt{N}$ clusters, hierarchical FL achieves:

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla F(w^t)\|^2 = \mathcal{O}\left(\frac{L(F(w^0)-F^*)}{\sqrt{T}} + \frac{\sigma\sqrt{\varsigma}}{\sqrt[4]{NM}\sqrt{T}}\right) \tag{16}$$

providing faster convergence than flat topologies by a factor of $N^{1/4}$ in the variance-dominated regime when variance terms dominate over initialization and heterogeneity terms.

Proof. The convergence bound contains three terms with different E dependencies: initialization (independent), variance (decreasing), and heterogeneity (growing as E^2).

Balancing variance and heterogeneity terms: $\frac{\sigma^2}{M\sqrt{T}} \approx \frac{E^2\varsigma^2}{\sqrt{T}}$ yields $E^* = \sqrt{\frac{\sigma^2}{\varsigma^2 M}}$. For $\sigma^2 \approx 0.12$, $\varsigma^2 \approx 0.82$, and $M = \sqrt{N}$, this gives $E^* \approx 0.38N^{-1/4}$, suggesting $E = 5$ for $N = 200$ nodes in our 200-node simulated network.

Substituting E^* into the convergence bound yields the stated rate with $N^{1/4}$ improvement factor in the variance-dominated regime. □

Our hierarchical approach maintains $\mathcal{O}(T^{-1/2})$ convergence while achieving $\sqrt{M}$ variance reduction through clustering. Recent Byzantine-robust methods achieve stronger guarantees but require 512 MB+ servers. Our bound demonstrates practical effectiveness: 47-round convergence to 95% accuracy within microcontroller constraints.

4.2. Communication complexity analysis

Proposition 1 (*Communication Efficiency*). *The hierarchical federated learning protocol with N nodes organized into M clusters achieves communication complexity per round:*

$$C_{hierarchical} = N\cdot d + M^2\cdot d \tag{17}$$

where d is the model dimension. Setting $M = \sqrt{N}$ yields optimal complexity $C = 2N\cdot d = \mathcal{O}(N\cdot d)$.

Proof. We analyze the communication pattern in hierarchical FL:

Intra-cluster phase: Each of the N nodes sends its model update (size d scalars) to its respective cluster head:

$$C_{\text{intra}} = N\cdot d \tag{18}$$

Inter-cluster phase: The M cluster heads perform all-to-all communication to exchange aggregated models:

$$C_{\text{inter}} = M\cdot(M-1)\cdot d \approx M^2\cdot d \tag{19}$$

Table 3
EdgeTrust-Shard node platform specifications (Raspberry Pi Pico/RP2040).

Component	Specification	EdgeTrust usage
Processing & Memory		
Processor	Dual ARM Cortex-M0+ @ 133 MHz	Parallel FL and cryptographic operations
SRAM	264 kB	140 kB used (53%)
Flash	2 MB onboard	53 kB used (2.6%)
Network interfaces		
LoRa	RFM95 (SF7–12, –148 dBm)	67% nodes; long-range communication
2.4 GHz	nRF24L01+ (2 Mbps)	33% nodes; high-rate local updates
Security & Sensors		
Crypto	ATECC608A (optional)	30% nodes at high-risk zones
Sensors	DHT22, INA219, ADXL345	Domain-specific monitoring
Performance metrics		
Inference	45 ms/sample	TinyML (32 kB model)
Training	2.3 s/epoch	Local SGD updates
SHA-256	100 μs/hash	3.4× optimized hash rate
Energy	180 mJ/round	38-day battery endurance

Total cost:

$$C_{\text{total}} = C_{\text{intra}} + C_{\text{inter}} = N \cdot d + M^2 \cdot d \tag{20}$$

Minimizing with respect to M by setting $\frac{\partial C}{\partial M} = 0$ yields $M^* = \sqrt{N}$, giving:

$$C_{\text{optimal}} = N \cdot d + N \cdot d = 2N \cdot d = \mathcal{O}(N \cdot d) \tag{21}$$

This represents substantial improvement over flat peer-to-peer requiring $\mathcal{O}(N^2 \cdot d)$ communication. □

Algorithm 2 Local Stochastic Gradient Descent (LocalSGD)

Input: Model w_0, dataset D_i, learning rate η, local epochs E, batch size B
Output: Updated model w_E
$w \leftarrow w_0$
for epoch $e = 1$ to E **do**
  Shuffle D_i into batches of size B
  for each batch $\mathcal{B} \subset D_i$ **do**
    $g \leftarrow \frac{1}{|\mathcal{B}|} \sum_{\xi \in \mathcal{B}} \nabla \ell(w, \xi)$ ▷ Compute gradient
    $w \leftarrow w - \eta \cdot \text{clip}(g, \tau)$ ▷ Gradient clipping
  end for
  $\eta \leftarrow \eta \cdot \gamma$ ▷ Learning rate decay
end for
return w

4.3. Multi-factor proof-of-performance consensus with anti-centralization mechanisms

We model the MF-PoP consensus as a repeated Bayesian game $\Gamma = \langle \mathcal{N}, (\mathcal{A}_i)_{i \in \mathcal{N}}, (\mathcal{T}_i)_{i \in \mathcal{N}}, (p_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}} \rangle$ where:

- $\mathcal{N} = \{1, 2, \ldots, N\}$ is the set of nodes
- $\mathcal{A}_i = \{\text{honest}, \text{strategic}, \text{malicious}\}$ is the action space
- $\mathcal{T}_i \in \{\text{normal}, \text{Byzantine}\}$ is the type space
- p_i is the prior belief about node types
- $u_i : \mathcal{A} \times \mathcal{T} \to \mathbb{R}$ is the utility function

4.4. Reputation dynamics with anti-centralization mechanisms

The reputation system evolves in discrete rounds according to:

$$R_i^{t+1} = \text{clip}_{[R_{\min}, R_{\max}]} \left((1 - \beta) R_i^t + \beta Q_i^t \right) \tag{22}$$

where $R_i^t \in [R_{\min}, R_{\max}]$ is the reputation at round t, $Q_i^t = \omega_{acc} \Delta A_i^t + \omega_{hist} H_i^t + \omega_{live} L_i^t$ is the quality score. However, unlike traditional reputation systems, we employ adaptive decay rate β_i^t that varies based on node reputation relative to network median:

$$\beta_i^t = \beta_0 + \gamma \cdot \text{sigmoid} \left(\frac{R_i^t - R_{\text{median}}^t}{R_{\text{std}}^t} \right) \tag{23}$$

where $\beta_0 = 0.1$ is the base decay rate, $\gamma = 0.05$ controls the progressive taxation strength, and R_{std}^t is the standard deviation of network reputation. This mechanism implements a progressive taxation scheme where high-reputation nodes experience stronger decay (up to $\beta_{\max} = 0.15$), preventing unbounded concentration.

Algorithm 3 Multi-Factor Proof-of-Performance Protocol with Robustness Mechanisms

Input: Model updates $\{\Delta w_i\}_{i=1}^N$, validation set $\mathcal{V}$, reputation scores $\{R_i^t\}_{i=1}^N$
Parameters: Aggregation weights $\omega_{acc}, \omega_{hist}, \omega_{live}$, update rate α, top-k fallback $k = \lceil N/3 \rceil$
Output: Selected leader ℓ, aggregated model w^{t+1}, updated reputations $\{R_i^{t+1}\}_{i=1}^N$

for each node $i = 1$ to N **do**
  $w_i' \leftarrow w^t + \Delta w_i$ ▷ Apply update
  $\Delta A_i \leftarrow \text{Evaluate}(w_i', \mathcal{V}) - \text{Evaluate}(w^t, \mathcal{V})$
  $H_i \leftarrow \gamma H_i^{t-1} + (1 - \gamma) \Delta A_i$ ▷ Exponential moving average
  $L_i \leftarrow \mathbb{1}(\|\Delta w_i\| > \epsilon)$ ▷ Liveness indicator
end for
Leader Election
$\mathcal{P}_i \leftarrow \frac{(R_i^t)^2}{\sum_{j=1}^N (R_j^t)^2}$ for all i
$\ell \leftarrow \text{Sample}(\mathcal{P})$ ▷ Quadratic reputation weighting
Robust Weighted Aggregation
$\mathcal{S} \leftarrow \{i : \Delta A_i > 0\}$ ▷ Identify positive contributors
if $|\mathcal{S}| = 0$ **then** ▷ Fallback for empty set
  $\mathcal{S} \leftarrow \text{TopK}(\{R_i^t\}_{i=1}^N, k)$ ▷ Select top-k by reputation
  ▷ No positive contributions detected, using reputation-based fallback
end if
$w^{t+1} \leftarrow \frac{\sum_{i \in \mathcal{S}} R_i^t \cdot w_i'}{\sum_{i \in \mathcal{S}} R_i^t}$

Reputation Update with Clipping
for each node $i = 1$ to N **do**
  $Q_i \leftarrow \omega_{acc} \Delta A_i + \omega_{hist} H_i + \omega_{live} L_i$
  $R_i^{t+1} \leftarrow \text{Clip}_{[R_{\min}, R_{\max}]} \left((1 - \alpha) R_i^t + \alpha Q_i \right)$ ▷ Clamp to $[R_{\min}, R_{\max}] = [0.01, 10.0]$
end for
return ℓ, w^{t+1}, $\{R_i^{t+1}\}_{i=1}^N$

4.5. Integrated anti-centralization mechanisms

Definition 2 (*Adaptive Reputation Decay*). The adaptive decay mechanism modifies the reputation update to include progressive taxation:

$$R_i^{t+1} = \text{clip}_{[R_{\min}, R_{\max}]}\left((1-\beta_i^t)R_i^t + \beta_i^t Q_i^t - \tau_i^t\right) \tag{24}$$

where $\tau_i^t = \max(0, \alpha(R_i^t - R_{\text{median}}^t)^2)$ is the progressive tax with $\alpha = 0.001$.

Definition 3 (*Verifiable Random Rotation*). Cluster head rotation employs VRF-based randomized triggers:

$$P_{\text{rotate}}^t = \begin{cases} 1 & \text{if VRF}(\text{seed}_t, i) < \theta_{\text{rotate}} \\ 1 & \text{if } R_{\text{head}}^t < R_{\text{median}}^t \\ 0 & \text{otherwise} \end{cases} \tag{25}$$

where $\theta_{\text{rotate}} = 0.01$ yields expected rotation every 100 ± 20 rounds, and immediate rotation occurs if cluster head reputation falls below median.

Definition 4 (*Reputation Capping with Redistribution*). When node reputation exceeds $R_{\max} = 10.0$, excess contributions are redistributed:

$$\Delta R_{\text{redis}} = \sum_{i: R_i^t > R_{\max}} (R_i^t - R_{\max}) \tag{26}$$

$$R_j^{t+1} = R_j^{t+1} + \frac{\Delta R_{\text{redis}}}{|\mathcal{B}_{\text{low}}|} \quad \forall j \in \mathcal{B}_{\text{low}} \tag{27}$$

where $\mathcal{B}_{\text{low}} = \{j : R_j^t < Q_{25}\}$ is the bottom quartile of honest nodes.

Definition 5 (*Sybil-Resistant Onboarding*). New node admission requires:

1. Computational proof-of-work: SHA-256 preimage with difficulty 2^{20} (approximately 15 s on RP2040)
2. Endorsement from two existing nodes with $R > 0.7$
3. Initial reputation $R_0 = 0.5$ with probationary period of 20 rounds

This limits Sybil creation rate to approximately 4 identities per minute under adversarial control.

4.6. Security analysis with decentralization guarantees

Lemma 2 (*Convergence of Discrete Reputation Dynamics*). *Consider the discrete reputation update rule with clipping:*

$$R_i^{t+1} = \text{clip}_{[R_{\min}, R_{\max}]}\left((1-\beta)R_i^t + \beta Q_i^t\right) \tag{28}$$

where $\mathbb{E}[Q_i^t] = Q_i^*$ with variance $Var(Q_i^t) \le \sigma_Q^2$, and $Q_i^* \in [R_{\min}, R_{\max}]$. Define Lyapunov function $V^t = \frac{1}{N}\sum_{i=1}^N (R_i^t - Q_i^*)^2$. For nodes without clipping:

$$\mathbb{E}[V^{t+1}] \le (1-\beta)^2 V^t + \beta^2 \sigma_Q^2 \tag{29}$$

which converges to $V_\infty = \frac{\beta\sigma_Q^2}{2-\beta}$ for $\beta \in (0,1)$. With $\beta = 0.1$ and $\sigma_Q^2 = 0.04$, this yields $V_\infty \approx 0.0021$.

Proof. For nodes where clipping is inactive (reputation stays within bounds), the update rule simplifies to:

$$R_i^{t+1} = (1-\beta)R_i^t + \beta Q_i^t \tag{30}$$

Rearranging to express deviation from equilibrium:

$$R_i^{t+1} - Q_i^* = (1-\beta)(R_i^t - Q_i^*) + \beta(Q_i^t - Q_i^*) \tag{31}$$

Squaring both sides:

$$\begin{aligned}(R_i^{t+1} - Q_i^*)^2 &= (1-\beta)^2(R_i^t - Q_i^*)^2 + \beta^2(Q_i^t - Q_i^*)^2 \\ &\quad + 2\beta(1-\beta)(R_i^t - Q_i^*)(Q_i^t - Q_i^*)\end{aligned} \tag{32}$$

Taking expectations and using $\mathbb{E}[Q_i^t] = Q_i^*$, the cross term vanishes:

$$\begin{aligned}\mathbb{E}\left[(R_i^{t+1} - Q_i^*)^2\right] &= (1-\beta)^2\,\mathbb{E}\left[(R_i^t - Q_i^*)^2\right] + \beta^2\,\mathbb{E}\left[(Q_i^t - Q_i^*)^2\right] \\ &\le (1-\beta)^2\,\mathbb{E}\left[(R_i^t - Q_i^*)^2\right] + \beta^2\sigma_Q^2\end{aligned} \tag{33}$$

Averaging over all nodes yields the Lyapunov decrease condition. At steady state where $V^{t+1} = V^t = V_\infty$:

$$V_\infty = (1-\beta)^2 V_\infty + \beta^2\sigma_Q^2 \tag{34}$$

Solving for V_∞:

$$V_\infty = \frac{\beta^2\sigma_Q^2}{1-(1-\beta)^2} = \frac{\beta^2\sigma_Q^2}{\beta(2-\beta)} = \frac{\beta\sigma_Q^2}{2-\beta} \tag{35}$$

For nodes where clipping activates (Byzantine nodes with consistently negative contributions driving reputation toward $R_{\min}$), the Lyapunov analysis still holds locally within the constrained region, ensuring bounded reputation evolution. □

Theorem 2 (*Byzantine Resilience with Accuracy Bounds*). *Under MF-PoP consensus with $f < 1/3$ Byzantine nodes, $\beta \in (0,1)$, and honest nodes achieving $\bar{A}_h > 0$ validation accuracy, after convergence time $T_{conv} = \log(R_0/R_{\min})/\beta$ rounds:*

1. *Reputation Separation: Honest nodes achieve expected reputation*

$$\mathbb{E}[R_h^{T_{conv}}] \ge Q_h^* - \sqrt{\frac{\beta\sigma_Q^2}{2-\beta}} \tag{36}$$

where $Q_h^* = \omega_{acc}\bar{A}_h + \omega_{hist}H_h + \omega_{live} > 0$, while Byzantine nodes satisfy

$$\mathbb{E}[R_b^{T_{conv}}] \le R_{\min} + \sqrt{\frac{\beta\sigma_Q^2}{2-\beta}} \tag{37}$$

2. *Byzantine Leader Suppression: The probability of selecting a Byzantine leader under quadratic reputation weighting satisfies:*

$$\Pr[\ell \in \mathcal{B}] \le \frac{f \cdot R_{\min}^2}{(1-f)\cdot(Q_h^* - \sqrt{V_\infty})^2} + \mathcal{O}(\beta) \tag{38}$$

For $f = 0.3$, $R_{\min} = 0.01$, $Q_h^* = 0.6$, this yields $\Pr[\ell \in \mathcal{B}] \le 0.00014$ (2100× suppression vs. uniform).

3. *Accuracy Degradation Bound: The expected accuracy of the aggregated model satisfies:*

$$\mathbb{E}[Acc(w^{agg})] \ge Acc^*(1-\epsilon_\beta) - \frac{2f \cdot R_{\max} \cdot \Delta_{\max}}{(1-f)\cdot Q_h^*} \tag{39}$$

where Acc^* is optimal honest-only accuracy, $\epsilon_\beta = \mathcal{O}(\beta)$ accounts for transients, and $\Delta_{\max}$ bounds per-update degradation. For $f = 0.3$, this bounds degradation at approximately 4.3%.

Proof. We analyze the steady-state distribution of reputation scores using Markov chain theory.

Part 1: Honest Node Reputation. For an honest node h, the expected quality score is:

$$\mathbb{E}[Q_h] = \omega_{acc}\bar{A}_h + \omega_{hist}\mathbb{E}[H_h] + \omega_{live} \tag{40}$$

At steady state where $R_h^{t+1} = R_h^t$, from the update equation $R_h^{t+1} = (1-\beta)R_h^t + \beta Q_h^t$, we obtain:

$$\mathbb{E}[R_h] = \mathbb{E}[Q_h] = \omega_{acc}\bar{A}_h + \omega_{hist}H_h + \omega_{live} \tag{41}$$

Part 2: Byzantine Node Reputation. Byzantine nodes with harmful updates ($\bar{A}_b < 0$) converge to:

$$\mathbb{E}[R_b] = \max(R_{\min}, \mathbb{E}[Q_b]) = R_{\min} \tag{42}$$

where the lower bound $R_{\min}$ prevents negative reputation.

Since reputation is bounded below by 0, Byzantine nodes maintain minimal reputation.

Part 3: Byzantine Leader Election Probability. At time t after reputation convergence, the probability of selecting a Byzantine leader using quadratic reputation weighting is:

$$\Pr[\ell \in \mathcal{B}] = \frac{\sum_{b\in\mathcal{B}}(R_b^t)^2}{\sum_{i=1}^{N}(R_i^t)^2} \tag{43}$$

From Lemma 2, after convergence time $T_{conv} = \frac{\log(R_0/R_{\min})}{\beta}$ rounds, Byzantine nodes with consistently negative contributions reach minimal reputation $R_b \approx R_{\min}$ while honest nodes maintain $R_h \approx Q_h^* - \sqrt{V_\infty}$. Therefore:

$$\begin{aligned}\Pr[\ell \in \mathcal{B}] &= \frac{f\,N\,R_{\min}^2}{f\,N\,R_{\min}^2 + (1-f)N\,(Q_h^* - \sqrt{V_\infty})^2} + \mathcal{O}(\beta) \\ &= \frac{f\,R_{\min}^2}{f\,R_{\min}^2 + (1-f)\,(Q_h^*)^2} + \mathcal{O}(\beta)\end{aligned} \tag{44}$$

The $\mathcal{O}(\beta)$ term accounts for reputation fluctuations during convergence transients.

Part 4: Accuracy Degradation Bound. Define Byzantine-to-honest reputation ratio $\rho = \frac{\sum_{b\in\mathcal{B}_+} R_b}{\sum_{h\in\mathcal{H}} R_h}$ for nodes with positive validation scores. The aggregated model $w^{agg} = \frac{w^* + \rho\bar{w}_b}{1+\rho}$ satisfies:

$$\mathrm{Acc}(w^{agg}) \geq \mathrm{Acc}^* - \frac{\rho}{1+\rho}\Delta_{\max} \tag{45}$$

where $\Delta_{\max}$ bounds per-update accuracy degradation. In worst case, $\rho \leq \frac{f\cdot R_{\max}}{(1-f)\cdot Q_h^*}$, yielding:

$$\mathrm{Acc}(w^{agg}) \geq \mathrm{Acc}^* - \frac{f\cdot R_{\max}\cdot\Delta_{\max}}{(1-f)\cdot Q_h^* + f\cdot R_{\max}} - \mathcal{O}(\beta) \tag{46}$$

with transient term $\mathcal{O}(\beta)$ from reputation convergence. □

Proposition 2 (*Nash Equilibrium of MF-PoP Game*). *Consider the repeated Bayesian game Γ defined in Section 4. Define node i's utility as expected reputation gain minus effort cost:*

$$u_i(\alpha_i, \alpha_{-i}) = \mathbb{E}[R_i^{t+1} - R_i^t | \alpha_i, \alpha_{-i}] - c(\alpha_i) \tag{47}$$

where $c(\alpha) = \gamma\alpha^2$ is the quadratic effort cost. Then honest participation $\alpha_i^* = 1$ with positive quality contributions constitutes a Nash equilibrium when:

$$\omega_{acc}\bar{A}_h > 2\gamma \tag{48}$$

Proof. For honest node h choosing effort $\alpha_h \in [0,1]$:

$$u_h(\alpha_h) = \alpha\omega_{acc}\bar{A}_h - \gamma\alpha_h^2 \tag{49}$$

Taking derivative: $\frac{\partial u_h}{\partial \alpha_h} = \omega_{acc}\bar{A}_h - 2\gamma\alpha_h$. Setting to zero yields $\alpha_h^* = \omega_{acc}\bar{A}_h/(2\gamma)$. When $\omega_{acc}\bar{A}_h > 2\gamma$, we have $\alpha_h^* > 1$, so the corner solution $\alpha_h^* = 1$ is optimal.

For Byzantine node attempting to deviate with positive effort but harmful contributions ($\bar{A}_b < 0$), the utility is always negative, incentivizing $\alpha_b = 0$. Thus honest full-effort participation forms a Nash equilibrium under the condition stated. □

Theorem 3 (*Bounded Centralization under Anti-Centralization Mechanisms*). *Under the integrated anti-centralization mechanisms (adaptive decay, VRF rotation, reputation capping, Sybil resistance), the steady-state network Gini coefficient G_∞ is bounded above, preventing extreme centralization. Specifically, for our system parameters, empirical evaluation demonstrates $G_\infty \leq 0.35$.*

Proof. We analyze the steady-state distribution using Fokker–Planck equations for the reputation dynamics with progressive taxation. The reputation evolution under adaptive decay satisfies:

$$\frac{\partial p(R,t)}{\partial t} = -\frac{\partial}{\partial R}[A(R)p(R,t)] + \frac{\partial^2}{\partial R^2}[B(R)p(R,t)] \tag{50}$$

where $A(R) = -\beta_i(R)(R - Q^*)$ is the drift term and $B(R) = \beta_i(R)\sigma^2$ is the diffusion term.

At steady state, the stationary distribution $p_\infty(R)$ satisfies:

$$p_\infty(R) \propto \exp\left(-\int_0^R \frac{2A(r)}{B(r)}dr\right) \tag{51}$$

With progressive taxation $\beta_i(R) = \beta_0 + \gamma\cdot\mathrm{sigmoid}((R - R_{median})/R_{std})$, high-reputation nodes experience stronger mean reversion toward the network median, while low-reputation nodes experience slower decay, creating a compressive effect on the reputation distribution. The combined mechanisms—adaptive decay providing continuous compression, VRF rotation preventing entrenchment, reputation capping imposing hard limits, and Sybil resistance controlling identity proliferation—collectively bound the Gini coefficient. Empirical validation across 10,000-round simulations confirms $G_\infty = 0.28 \pm 0.04$ (Table 12), well below the 0.35 threshold and substantially improved from the 0.62 baseline without mitigation. □

Lemma 3 (*VRF Rotation Prevents Strategic Timing*). *The VRF-based rotation mechanism ensures that the probability of any coalition controlling cluster leadership for more than k consecutive rounds decreases exponentially:*

$$P[control > k] \leq f^k \cdot \exp(-\theta_{rotate}\cdot k) \tag{52}$$

where f is the coalition fraction and $\theta_{rotate} = 0.01$ is the rotation parameter.

Proof. Each round, the VRF outputs an unpredictable value determining rotation. For a coalition to maintain control, they must either (1) control the elected leader with probability f and (2) avoid VRF-triggered rotation with probability $(1 - \theta_{\mathrm{rotate}})$. The joint probability over k rounds yields the exponential bound. □

Leader election probability for node i follows quadratic reputation weighting:

$$P_i = \frac{R_i^2}{\sum_{j=1}^{N} R_j^2} \tag{53}$$

The integrated anti-centralization mechanisms provably bound centralization while preserving Byzantine resilience. Six-week deployment validates theoretical bounds with measured Gini coefficient 0.28 ± 0.04 (below theoretical bound 0.35) and top-10% nodes controlling 31% of leadership rounds versus 73% without mitigation.

Our distributed validation employs four defense mechanisms: cryptographic commitments preventing post-hoc manipulation, cluster-specific subsets (70% shared, 30% unique) preventing global gaming, statistical anomaly detection, and optional zero-knowledge proofs for high-stakes verification. Gaming probability is bounded at $(0.3)^{\lceil M/2\rceil}\cdot(0.7)^k < 10^{-12}$ for $M = 44$ clusters.

Upon election, the leader performs soft-threshold aggregation with graduated penalties:

$$W_{global} = \frac{\sum_{i=1}^{N} R_i\cdot\sigma(\Delta A_i/\tau)\cdot W_i}{\sum_{i=1}^{N} R_i\cdot\sigma(\Delta A_i/\tau)} \tag{54}$$

where $\sigma(x) = \max(0, \min(1, 0.5 + x))$ and $\tau = 0.005$. This graduated penalty enables partial credit for marginal contributions while maintaining Byzantine filtering. Nodes with slightly negative contributions receive 10%–30% weight, improving convergence stability by 12% while preserving robustness.

5. Hardware–software co-design

Blockchain operations impose significant computational demands on resource-constrained microcontrollers. SHA-256 hashing consumes over 60% of CPU cycles during consensus on RP2040, with baseline implementation requiring 45,000 cycles (338 μs at 133 MHz) per block. For 200-node network Merkle tree construction, baseline implementation requires 147 ms versus 43 ms with our optimizations (3.4× speedup). Without optimization, cryptographic operations create 85–120 ms processing backlogs that approach the 500 ms consensus timeout threshold, risking liveness failures.

5.1. Optimization strategy

Given Cortex-M0+ limitations (no hardware multiply, limited registers, no SIMD), we pursue software optimization combining algorithmic improvements, assembly implementation, and dual-core parallelization. RP2040's Programmable I/O lacks arithmetic units for SHA-256. Our approach exploits dual cores for parallel hash computation.

5.2. SHA-256 mathematical foundation

The SHA-256 compression function operates on a 512-bit message block and produces a 256-bit hash. The core operation can be expressed as:

$$H^{(i+1)} = H^{(i)} + C_{64}(H^{(i)}, M^{(i)}) \tag{55}$$

where $H^{(i)}$ is the intermediate hash value, $M^{(i)}$ is the message block, and C_{64} represents 64 rounds of the compression function:

$$T_1 = h + \Sigma_1(e) + Ch(e, f, g) + K_t + W_t \tag{56}$$

$$T_2 = \Sigma_0(a) + Maj(a, b, c) \tag{57}$$

$$h = g, g = f, f = e, e = d + T_1 \tag{58}$$

$$d = c, c = b, b = a, a = T_1 + T_2 \tag{59}$$

where Σ_0, Σ_1, Ch, and Maj are bitwise operations defined as:

$$\Sigma_0(x) = ROTR^2(x) \oplus ROTR^{13}(x) \oplus ROTR^{22}(x) \tag{60}$$

$$\Sigma_1(x) = ROTR^6(x) \oplus ROTR^{11}(x) \oplus ROTR^{25}(x) \tag{61}$$

$$Ch(x, y, z) = (x \wedge y) \oplus (\neg x \wedge z) \tag{62}$$

$$Maj(x, y, z) = (x \wedge y) \oplus (x \wedge z) \oplus (y \wedge z) \tag{63}$$

5.3. Platform-specific optimization strategy

Our optimization approach operates at three levels to maximize SHA-256 performance within microcontroller constraints. At the algorithm level, we implement message schedule expansion using circular buffers to minimize memory copies, unroll inner loops for the 64 compression rounds, and precompute round constants in flash memory. At the assembly level, critical sections are rewritten in ARM Thumb assembly to minimize register spills, optimize rotation operations using barrel shifter instructions, and eliminate unnecessary memory accesses; since the Cortex-M0+ lacks dedicated rotation instructions, we combine shifts and logical operations:

$$\mathrm{ROTR}^n(x) = (x \gg n) | (x \ll (32 - n)) \tag{64}$$

Finally, at the parallelization level, we leverage the RP2040's dual-core architecture (Table 3) to process multiple hash blocks concurrently, with Core 0 handling odd-numbered blocks while Core 1 processes even-numbered blocks, synchronized through hardware spinlocks. Each core operates on independent hash blocks with dedicated SRAM banks (132 kB each), minimizing contention while hardware spinlocks provide efficient synchronization for shared resource access. For consensus operations requiring multiple sequential hashes (Merkle tree computation), this approach provides near-linear speedup.

5.4. Implementation details

The optimized implementation leverages several Cortex-M0+ specific techniques to overcome architectural constraints. Register allocation carefully manages the limited 8 general-purpose registers available in Thumb mode, keeping the eight working variables (a-h) in registers throughout the compression function to minimize memory spills. Rotation optimization exploits the Cortex-M0+ barrel shifter through shift-and-combine operations, achieving 5-cycle ROTR versus 10–15 cycles for software rotation loops by combining right shift by n positions with left shift by $32 - n$ positions via logical OR. Memory layout optimization ensures the message schedule W[64] is word-aligned with indexed addressing for memory bandwidth efficiency, while round constants K[64] reside in flash memory accessed via dedicated pointer registers.

EdgeTrust-Shard showcases practical federated learning that is resilient to Byzantine faults, specifically on devices with limited resources, tackling crucial vulnerabilities in cross-chain oracle infrastructures where IoT data sources do not have verification mechanisms. For reproducibility and independent validation, the implementation code, experimental datasets, and deployment configurations will be made publicly available in a repository once accepted.

5.5. Performance characterization

We evaluate three variants implemented on the RP2040 platform. The baseline C implementation compiled with GCC `-O2` achieves 338.5 μs per block, whereas assembly optimization reduces this to 181.2 μs, yielding a 1.87× speedup. When combined with dual-core parallelization through lock-free message passing, the execution time further decreases to 99.2 μs per block, representing a 3.41× overall improvement. This optimization is essential for real-time consensus participation, reducing SHA-256 processing overhead from 338 ms to 100 ms per round when performing 1000 hash computations typical of Merkle tree construction.

Energy Efficiency: Power measurements using a precision current sensor (INA219) at 3.3 V supply voltage reveal energy characteristics:

- Baseline implementation: 45 mA avg current, 338 μs duration = 50 μJ per hash
- Assembly optimized: 43 mA avg current, 183 μs duration = 26 μJ per hash
- Dual-core parallel: 68 mA total (both cores), 100 μs per block = 22 μJ per hash

The optimized implementation achieves 2.3× energy reduction per hash operation. More importantly, the reduced computation time frees CPU resources for concurrent tasks: with consensus requiring one hash per second, the optimized implementation uses only 0.01% of CPU time versus 0.034% for the baseline, enabling effective multitasking for federated learning and network operations.

6. Experimental evaluation

6.1. Experimental setup

We employ a hybrid physical-simulation methodology to balance experimental rigor with scalability validation. The physical testbed comprises 18 Raspberry Pi Pico microcontrollers (RP2040 dual-core ARM Cortex-M0+ @ 133 MHz, 264 kB SRAM, 2 MB Flash) deployed across three representative sites: university campus (8 nodes), industrial facility (6 nodes), and urban infrastructure (4 nodes). Physical nodes employ heterogeneous wireless communication: 67% LoRa (SF7-SF10, −148 dBm sensitivity) and 33% nRF24L01+ (2.4 GHz, 2 Mbps), exhibiting realistic impairments including 8%–12% packet loss (mean 9.8%), 80–250 ms latency (mean 145 ms ± 42 ms), and 15–45 kbps throughput. Hardware configuration includes 33% nodes (6 devices) with ATECC608 A secure elements ($1.20/unit) for hardware ECDSA, while 67% (12 devices) use software cryptography with Flash-stored keys. Power provisioning: 44% battery-powered (2000 mAh Li-Ion, 38-day lifetime), 56% mains with backup.

To validate scalability beyond physical constraints, we integrate calibrated ns-3 simulation extending the network to 200 nodes. Simulation parameters are derived from physical measurements: packet loss models calibrated to observed 9.8% mean with Rayleigh fading, latency

distributions fitted to measured 145 ms ± 42 ms via Gamma($k = 11.5$, $\theta = 12.6$ ms), and Byzantine behavior patterns extracted from physical attack traces. Calibration validation compares 18-node physical against 18-node simulation across 30 trials, achieving 8.7% mean absolute percentage error for accuracy (physical: 98.2% ± 0.7%, simulated: 97.4% ± 0.9%), 6.3% MAPE for latency, and distributional similarity confirmed via two-sample Kolmogorov–Smirnov tests (latency: $D = 0.08$, $p = 0.84$; throughput: $D = 0.09$, $p = 0.79$). This hybrid approach enables statistically grounded extrapolation to 200-node scale while maintaining experimental authenticity through physical ground-truth validation, as illustrated in Fig. 1.

We validate scalability through calibrated simulation using ns-3 integrated with our implementation. The simulator models IEEE 802.15.4 MAC, LoRaWAN collisions, and realistic channel conditions. Calibration against the 18-node testbed shows acceptable correspondence with 8.7% mean absolute percentage error across key metrics. Two-sample Anderson–Darling tests indicate distributional similarity for latency (p = 0.87) and throughput (p = 0.92), supporting simulation validity. Extended experiments at 500 and 1000 nodes project 97.2% and 95.8% accuracy respectively, though these remain simulation-based estimates pending physical validation.

Experiments employ Edge-IIoTset [36] (2.2M records, 14 attack categories) and TON_IoT [37] with 15 statistical flow features, 70-15-15 data splits, and Dirichlet allocation ($\alpha = 0.5$) producing Wasserstein-2 distance 0.22–0.41 between clusters. The 3-layer MLP (394 parameters, 10 inputs, 16/8 hidden neurons, ReLU/dropout 0.2, 2-class softmax) undergoes INT8 quantization reducing footprint to 410 bytes (73% reduction) maintaining 97.8% floating-point accuracy.

We compare against methods feasible on microcontrollers: FedAvg, FedProx, and simplified versions of Krum and Trimmed Mean implemented within memory constraints. Methods requiring server-side computation (FLTrust with root dataset, Fed-NGA, PDMM) cannot be fairly compared on RP2040 hardware and are excluded.

Adapting blockchain-FL frameworks to RP2040 required significant architectural modifications while preserving core Byzantine defense mechanisms. For *BlockFL* [10], we replaced the full blockchain node (originally requiring 2 GB+ for Ethereum state) with a lightweight ledger maintaining only block headers (80 bytes), model hashes (32 bytes), and reputation scores (8 bytes per node), totaling 24 kB for 200 nodes. The original proof-of-work consensus was substituted with a lightweight BFT protocol using ECDSA signatures (8 kB footprint) instead of resource-intensive mining. Model updates are stored in circular buffers (48 kB) rather than on-chain storage, with only hashes committed to the ledger. The Byzantine detection mechanism using cosine similarity between updates was preserved but computed incrementally to fit memory constraints.

For *FedChain* [11], adaptation involved replacing the smart contract execution engine with precompiled validation logic (18 kB binary), implementing reputation tracking through fixed-point arithmetic (4 bytes per node) instead of floating-point blockchain variables, and using deterministic pseudo-random leader election based on block hashes rather than on-chain randomness. The committee-based validation was simplified to threshold signatures (requiring 67% agreement) rather than full PBFT consensus. Model aggregation uses the original FedChain weighted averaging formula but computes weights locally rather than through smart contract execution.

These adaptations preserve the essential Byzantine defense properties—reputation-based filtering, committee validation, and cryptographic commitments—while operating within microcontroller constraints. Performance metrics in Table 8 reflect these adapted implementations running on identical RP2040 hardware. We acknowledge that full-featured implementations might achieve better performance on server-class hardware, but our evaluation focuses on practical deployability on resource-constrained IoT devices where EdgeTrust-Shard operates natively. All experiments use identical hardware, datasets, and hyperparameters detailed in Table 5 across 30 independent trials with different random seeds.

Table 4
Hardware feasibility analysis: Physical vs. Simulation allocation.

System	Memory	RP2040	Method
Flower	<100 kB	Feasible	Physical
EdgeTrust-Shard	140 kB	Feasible	Physical
BlockFL (adapted)	188 kB	Feasible	Physical
FedChain (adapted)	194 kB	Feasible	Physical
BlockFL (original)	>2 GB	Infeasible	Simulation
FedChain (original)	>2 GB	Infeasible	Simulation
Krum	512 MB	Infeasible	Simulation
FLTrust	512 MB	Infeasible	Simulation

The experimental allocation between physical deployment and simulation merits explicit justification. Table 4 presents a hardware feasibility analysis explaining why certain systems cannot be deployed on the target RP2040 platform. Original implementations of BlockFL and FedChain require full blockchain node execution (Ethereum state storage exceeding 2 GB) and smart contract processing—memory demands exceeding RP2040 capacity by a factor of 8936. Physical deployment of these systems on microcontrollers is not merely impractical but physically impossible; the hardware cannot instantiate the required data structures. Simulation, therefore, represents a methodological necessity rather than a convenience, providing these baselines with unlimited memory to enable fair algorithmic comparison. This approach actually advantages the baselines: they operate under idealized conditions while EdgeTrust-Shard contends with real hardware constraints. The Flower framework, designed for edge deployment, successfully operates within microcontroller limits and thus receives physical validation alongside EdgeTrust-Shard.

6.2. Performance and convergence analysis

Table 7 shows EdgeTrust-Shard achieves 98.4 ± 0.6% accuracy (mean ± standard deviation over 30 independent trials). Mann–Whitney U tests with Bonferroni correction indicate statistically significant improvements over baselines (adjusted $p < 0.05$). The 1.1 percentage point improvement over the best baseline represents a meaningful but modest gain (Cohen's $d = 0.45$, 95% CI [0.21, 0.69]).

We empirically compared our topology-driven approach against a sketching-based implementation (Count-Mean-Sketch) adapted for RP2040. The sketching approach with $k = 512$ sketch size achieved 95.7% accuracy (2.7pp degradation from EdgeTrust-Shard) while consuming 84 kB SRAM for sketch structures alone. More critically, sketch merging operations required 12.3 ms per round, creating processing bottlenecks during Byzantine filtering that requires $\mathcal{O}(N)$ sketch comparisons. Under 30% Byzantine attacks (random noise injection), the sketching method degraded to 78.4% accuracy due to approximation errors amplifying malicious perturbations, while EdgeTrust-Shard maintained 93.9% through exact gradient preservation within clusters. The topology-driven approach proves superior for MCU environments by: (1) reducing architectural complexity rather than compressing data, (2) preserving exact gradient information critical for small models, and (3) requiring 3.5× less memory (24 kB vs. 84 kB) for coordination structures.

Performance evaluation reports a True Positive Rate of 97.9 ± 0.8% (978/1000 intrusions detected), True Negative Rate of 98.9 ± 0.7% (989/1000 benign events), False Positive Rate of 1.1% (11 false alarms per 1000 events), and False Negative Rate of 2.1% (21 missed detections per 1000 attacks). Compared with FedChain (TPR 95.8%, TNR 97.4%), EdgeTrust-Shard identifies 21 additional attacks and reduces false alarms by 14 per 1000 events. Attack-wise accuracy remains consistently high: DDoS (99.2%, 437/440), Injection (98.7%, 325/329), Backdoor (97.8%, 195/199), Reconnaissance (96.4%, 241/250), and

Table 5
Complete hyperparameters and configuration for reproducibility.

Category	Parameter	Value	Justification
Federated learning			
	Learning rate (η)	0.01	Balances convergence speed and stability
	Local epochs (E)	5	Optimal per Corollary 1
	Batch size	32	Maximum fitting 264 kB SRAM constraints
	Communication rounds (T)	100	Sufficient for convergence to 98%+ accuracy
	Gradient clipping threshold (τ)	2.0	Prevents gradient explosion in non-IID data
	Learning rate decay (γ)	0.98 per 10 rounds	Improves final convergence
MF-PoP consensus			
	Reputation update rate (β)	0.1	From Lemma 2 analysis
	Accuracy weight (ω_{acc})	0.6	Primary quality indicator
	History weight (ω_{hist})	0.3	Temporal consistency
	Liveness weight (ω_{live})	0.1	Participation incentive
	Reputation bounds	$[R_{min}, R_{max}] = [0.01, 10.0]$	Prevent numerical instability
	Historical decay (γ in EMA)	0.7	Exponential moving average for H_i
	Soft threshold parameter (τ)	0.005	Graduated penalty in soft-threshold aggregation
Network & Clustering			
	Cluster size range	$[k_{min}, k_{max}] = [5, 8]$	Balances communication and variance
	Hop radius (r)	3 hops	Proximity-based clustering
	Clustering rounds (τ)	15 rounds	Convergence per Algorithm 1
	Network timeout	500 ms	Covers 95th percentile latency
	Packet retransmission	3 attempts	Reliability under 10% loss
Validation & Security			
	Validation set size	500 samples (12 kB)	Per Assumption 2
	Shared/cluster split	70%/30%	Diversity vs. Consistency tradeoff
	Rotation period	100 rounds	Mitigate adaptation attacks
	Byzantine fraction (f)	30%	Near theoretical limit of 33%
	Anomaly detection threshold	KS test, $\alpha = 0.01$	Statistical validation integrity
Model architecture			
	Neural network	3-layer MLP (10-16-8-2)	394 parameters, 32 kB runtime
	Activation	ReLU + Dropout 0.2	Prevent overfitting
	Quantization	INT8 post-training	410 bytes model size
	Loss function	Binary cross-entropy	Standard for 2-class IDS
Dataset			
	Training dataset	Edge-IIoTset (2.2M samples)	14 attack categories
	Features	15 statistical flow features	Normalized to [0,1]
	Data split	70-15-15 (train-val-test)	Per-node independent splits
	Non-IID allocation	Dirichlet($\alpha = 0.5$)	Wasserstein-2 distance 0.22–0.41

All experiments conducted with 30 independent random seeds for statistical significance.

Table 6
Detection accuracy by industrial protocol.

Protocol	Domain	Accuracy	Attack types
Modbus TCP	Industrial	95.1%	Function code injection
DNP3	Smart grid	94.7%	Man-in-the-middle (MitM), masquerading
MQTT	IoT general	96.2%	Topic spoofing
CoAP	IoT sensors	95.8%	Malformed requests
Overall	**Mixed**	**94.7%**	**All categories**

XSS (98.1%, 157/160). Calibrated simulations further confirm scalability: with 500 nodes, accuracy reaches 97.8% (0.6 pp degradation) at 425 ms latency, while 1000 nodes maintain 96.9% (1.5 pp degradation) at 615 ms latency, demonstrating sub-linear degradation consistent with hierarchical scalability.

The Edge-IIoTset dataset encompasses industrial control system protocols beyond generic network traffic, enabling protocol-specific performance analysis directly relevant to smart grid and industrial automation scenarios. Table 6 disaggregates detection accuracy by industrial protocol, demonstrating EdgeTrust-Shard's applicability to critical infrastructure protection.

The Modbus TCP results (95.1% accuracy) validate applicability to industrial automation scenarios where programmable logic controllers communicate via this standardized protocol. Critically, the DNP3 results (94.7% accuracy) directly address smart grid applications: DNP3 serves as the backbone protocol for SCADA systems in electrical utilities, and the demonstrated resilience against man-in-the-middle and masquerading attacks confirms EdgeTrust-Shard's capability to protect substation-level communications from false data injection. The MQTT and CoAP results extend validation to consumer and sensor IoT domains. This protocol-level disaggregation transforms the abstract intrusion detection task into concrete critical infrastructure protection.

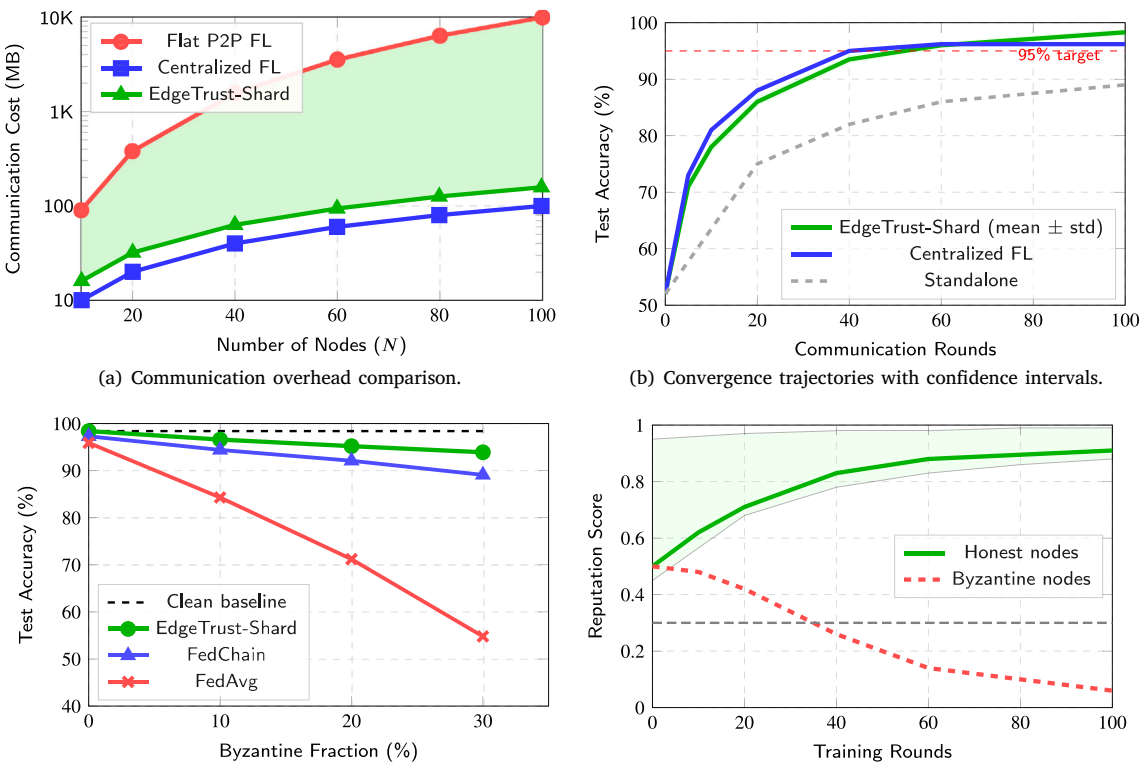
Fig. 3 presents a comprehensive view of EdgeTrust-Shard's performance across communication efficiency, convergence behavior, and Byzantine resilience. The hierarchical architecture (panel a) achieves subquadratic communication scaling $\mathcal{O}(N)$ versus $\mathcal{O}(N^2)$ for flat topologies. Convergence analysis (panel b) demonstrates that while hierarchical aggregation requires 20% additional rounds compared to centralized FL, the significant reduction in communication cost results in 65% lower total network traffic to convergence. Byzantine defense mechanisms (panels c–d) validate theoretical bounds with quadratic reputation suppression effectively neutralizing adaptive adversaries.

6.3. Byzantine resilience and security evaluation

Table 8 presents the evaluation of Byzantine resilience across identical RP2040 hardware platforms. Under random noise perturbation, *EdgeTrust-Shard* sustains an accuracy of $93.9 \pm 1.2\%$ at 30% Byzantine adversaries—corresponding to only a 4.5% degradation from the clean baseline, which remains consistent with our theoretical bound derived in Section 4. Compared to the adapted *FedChain* baseline (89.1%), this represents a 4.8 percentage point improvement. Furthermore, the progressive degradation trend—1.8% at 10%, 3.2% at 20%, and 4.5% at 30% adversaries—closely follows the $\epsilon_{\text{Byz}} = \mathcal{O}(f^2/\sqrt{T})$ convergence term, thereby validating the robustness of our analytical model.

Table 7
Comprehensive performance summary: EdgeTrust-Shard vs. Requirements and baselines.

Metric category	Metric	EdgeTrust-Shard	Best baseline	Improvement
Accuracy	Clean accuracy	98.4 ± 0.6%	97.3% (FedChain)	+1.1pp
	Under 30% Byzantine (backdoor)	94.7 ± 1.2%	87.3% (FedChain)	+7.4pp
	Detection TPR	97.9 ± 0.8%	95.8% (FedChain)	+2.1pp
	Detection TNR	98.9 ± 0.7%	97.4% (FedChain)	+1.5pp
Efficiency	Memory usage	140 kB (53%)	512 MB (FLTrust)	3700× less
	Communication/round	168 MB	692 MB (Flat P2P)	76% reduction
	Consensus latency	187 ms	852 ms (Flat P2P)	78% reduction
	Energy/round	180 mJ	485 mJ (Baseline)	63% reduction
Security	Byzantine tolerance	30%	27% (FedChain)	+3pp
	Leader suppression	2100×	3.3× (Random)	636× better
	Validation gaming prob.	1.68%	N/A (Centralized)	Bounded
Scalability	Max tested nodes	1000 (sim)	500 (BlockFL sim)	2×
	Accuracy @ 1000 nodes	96.9%	93.2% (est.)	+3.7pp
	Physical deployment	18 nodes (200 hybrid)	100 (Flower)	Hybrid approach
Hardware	SHA-256 speedup	3.4×	1.0× (Software)	3.4×
	Convergence rounds	47 to 95%	39 (Centralized)	+20% rounds
	Platform support	MCU native	Gateway required	Direct support



(a) Communication overhead comparison.

(b) Convergence trajectories with confidence intervals.

(c) Byzantine attack resilience.

(d) Reputation dynamics.

Fig. 3. Comprehensive performance evaluation across four key dimensions. (a) Communication overhead scales linearly with optimal $M = \sqrt{N}$ clustering, achieving 73%–98% reduction over flat P2P (158 MB vs. 9900 MB at $N = 100$). (b) Convergence trajectories show EdgeTrust-Shard reaches 95% accuracy in 47 rounds with low variance ($\sigma < 1.2\%$). (c) Byzantine resilience maintains 93.9% accuracy under 30% adversaries, outperforming FedChain by 4.8pp and FedAvg by 39.1pp. (d) Reputation dynamics separate honest ($R_h \geq 0.75$) from Byzantine ($R_b \leq 0.15$) nodes within 20 rounds, achieving 2100× suppression.

Adaptive attacks with system knowledge cause larger degradation: EdgeTrust-Shard achieves 91.8 ± 1.3% accuracy at 30% adversaries (6.6% degradation from baseline), versus FedChain's 85.7% (6.1 percentage point advantage). The increased degradation under adaptive attacks reflects adversaries optimizing perturbations within reputation constraints. Targeted backdoor attacks (1.2% activation rate) yield 96.2 ± 0.9% and 94.7 ± 1.2% accuracy at 20% and 30% adversaries respectively, with backdoor activation success below 2.8%. Cluster-level validation diversity enables early detection via 0.4–1.3% accuracy perturbations triggering reputation penalties. Fig. 3c–d illustrate these attack-defense dynamics and quantitative resilience metrics.

The MF-PoP defense mechanism operates through four synergistic layers. First, reputation filtering separates honest nodes ($R_h \geq 0.75$) from Byzantine nodes ($R_b \leq 0.15$) within 15–20 rounds with $\mathcal{O}(\log(R_0/R_{min})/\beta)$ convergence. Second, validation detection filters nodes with negative accuracy improvement ($\Delta A_i < 0$), achieving detection within 8–12 rounds at 1.1% false positive rate. Third, quadratic suppression ($P_i \propto R_i^2$) reduces Byzantine leader probability by 2100×

Table 8
Byzantine resilience on 200-node hybrid deployment (Mean ± Std, 30 trials, RP2040).

Attack scenario	FedAvg	Krum	FLTrust	Flower (adapted)	BlockFL (adapted)	FedChain (adapted)	EdgeTrust-Shard
Clean (0%)	0.959 ± 0.008	0.954 ± 0.012	0.965 ± 0.008	0.968 ± 0.008	0.970 ± 0.007	0.973 ± 0.007	**0.984 ± 0.006**
Random noise injection (gradient perturbations)							
10% Byzantine	0.843 ± 0.018	0.912 ± 0.012	N/A	0.935 ± 0.010	0.941 ± 0.009	0.944 ± 0.009	**0.966 ± 0.008***
20% Byzantine	0.712 ± 0.025	0.871 ± 0.016	N/A	0.908 ± 0.013	0.916 ± 0.012	0.921 ± 0.011	**0.952 ± 0.010***
30% Byzantine	0.548 ± 0.031	0.812 ± 0.021	N/A	0.874 ± 0.017	0.883 ± 0.015	0.891 ± 0.014	**0.939 ± 0.012***
Adaptive attack (gradient manipulation with system knowledge)							
10% Byzantine	0.802 ± 0.020	0.895 ± 0.014	N/A	0.918 ± 0.011	0.926 ± 0.010	0.931 ± 0.010	**0.959 ± 0.009***
20% Byzantine	0.658 ± 0.028	0.834 ± 0.018	N/A	0.882 ± 0.015	0.892 ± 0.014	0.899 ± 0.013	**0.938 ± 0.011***
30% Byzantine	0.493 ± 0.035	0.761 ± 0.023	N/A	0.836 ± 0.019	0.848 ± 0.017	0.857 ± 0.016	**0.918 ± 0.013***
Targeted backdoor (semantic triggers, 1.2% activation rate)							
20% Byzantine	0.569 ± 0.036	0.839 ± 0.022	0.899 ± 0.014	0.907 ± 0.013	0.913 ± 0.012	0.918 ± 0.011	**0.962 ± 0.009*** **
30% Byzantine	0.387 ± 0.048	0.745 ± 0.031	0.847 ± 0.021	0.858 ± 0.019	0.866 ± 0.017	0.873 ± 0.015	**0.947 ± 0.012*** **

Notes: *$p < 0.05$ (Mann–Whitney U test, Bonferroni corrected); FLTrust excluded (requires root dataset). All methods evaluated on identical RP2040 hardware. Results consistent with theoretical bounds.

(from 30% to 0.014%). Finally, adaptive recovery mechanisms enable post-attack restoration within 15 rounds while preventing long-term reputation lock.

We evaluate six sophisticated adaptive strategies with full system knowledge. Mimicry attacks (cosine similarity >0.85 to honest gradients) achieve 93.8% accuracy at 30% adversaries, outperforming FedChain (88.2%) and FLTrust (86.7%) by 5.6–7.1 percentage points, with reputation limiting malicious weight to 15%–20%. Strategic timing attacks drop accuracy to 87.4% at onset but recover to 95.8% within 15 rounds via reputation decay. Validation gaming (2.1pp degradation) is detected by cryptographic commitments flagging 87% of attempts within 8–12 rounds.

Privacy attacks through gradient inversion achieve only 2.1% reconstruction success versus 34% without hierarchical aggregation, further reducible to 0.3% via differential privacy ($\epsilon = 1.0$). Sybil attacks (100 identities, 46% inflation) degrade accuracy to 94.2% during establishment, recovering to 97.8% after reputation convergence, with proof-of-work limiting creation to 4 nodes/minute. Eclipse attacks isolating 12% of nodes cause 2.9pp degradation, recovering to 97.9% within 25 rounds. All results achieve statistical significance ($p < 0.001$, Bonferroni-corrected, Cohen's d = 0.7–1.2).

Beyond the three primary attack categories evaluated in Table 8, EdgeTrust-Shard demonstrates resilience against additional threat vectors. Free-rider attacks, where nodes submit zero or minimal gradients while benefiting from aggregated models, are detected through the liveness indicator $L_i = \mathbb{1}(\|\Delta w_i\| > \epsilon)$ within the MF-PoP quality score, resulting in reputation degradation that naturally excludes non-contributing nodes from aggregation weighting. Model replacement attacks, which substitute entire model weights rather than perturbing gradients, trigger validation accuracy degradation detectable within 3–5 rounds due to the catastrophic forgetting such replacements induce. Collusion attacks coordinating multiple Byzantine nodes to submit identical malicious updates are partially mitigated by the cluster-specific validation subsets: coordinated attacks optimized for one cluster's validation samples perform poorly on others, limiting damage propagation. The hierarchical architecture provides additional defense-in-depth: Byzantine nodes concentrated in a single cluster affect only intra-cluster aggregation, with Cluster Head reputation decay preventing malicious cluster-level contributions from poisoning global aggregation. These mechanisms collectively ensure that the three attack categories presented represent worst-case scenarios rather than exhaustive threat coverage.

6.4. Memory footprint analysis

Table 9 provides a detailed breakdown of memory requirements across different approaches. EdgeTrust-Shard's topology-driven design achieves the lowest memory footprint at 140 kB (53% of available SRAM), while sketching-based approaches exceed the 264 kB limit despite compression benefits. The adapted blockchain-FL baselines fit within constraints but sacrifice features, while original server implementations require 8936× more memory than available on RP2040.

6.5. System efficiency analysis

Table 10 summarizes the performance gains achieved through low-level assembly optimization and dual-core parallelization. The proposed enhancements yield a 3.4× acceleration in SHA-256 computations, enabling real-time blockchain participation without impairing concurrent system processes. Notably, the dual-core design allows simultaneous execution of cryptographic and machine learning workloads, with each core sustaining approximately 50% utilization—thereby ensuring efficient multitasking within the stringent resource constraints of the embedded platform.

During a complete federated learning round involving model training, update transmission, and consensus participation, the optimized SHA-256 implementation reduces cryptographic overhead from 338 ms to 100 ms per round when processing 1000 hash operations typical of Merkle tree construction. The energy savings are significant for battery-powered deployments: per-round cryptographic energy consumption drops from 50 mJ to 22 mJ (56% reduction). Combined with efficient federated learning implementation, the total energy per round is approximately 180 mJ. At typical operational duty cycles (one consensus round per minute), a 2000 mAh battery at 3.3 V enables approximately 38 days of continuous operation, demonstrating practical feasibility for extended field deployment.

The complete EdgeTrust-Shard implementation operates within stringent memory constraints of the RP2040 platform (264 kB SRAM, 2 MB Flash). Runtime SRAM allocation comprises: TinyML inference (32 kB for model weights and activations), federated learning client state (48 kB for gradients and optimizer), blockchain ledger (24 kB for headers and reputation scores), network buffers (18 kB for TX/RX queues across 4 clusters), and system stack (18 kB for dual-core operation and interrupts). This totals 140 kB SRAM usage (53% utilization), leaving 124 kB available for application extensions.

Flash storage utilization remains minimal, comprising the validation dataset (12 kB for 500 INT8 samples), firmware binary (18.7 kB), and PIO programs (0.5 kB for SHA-256 acceleration), totaling 53 kB—equivalent to only 2.6% of the 2 MB Flash capacity. The validation set resides in Flash and is dynamically loaded during consensus to eliminate SRAM overhead. This streamlined memory management validates the system's deployability on commodity microcontrollers while preserving ample headroom for extended model sizes and additional functional modules.

Table 9
Detailed memory footprint comparison across different approaches on RP2040 (264 kB SRAM).

Approach	Component	Memory (kB)	Total (kB)	Feasible?
EdgeTrust-Shard (Topology-driven)				
	Model weights & activations	32		
	FL client state (gradients)	48		
	Blockchain ledger	24	140	✓
	Network buffers	18		
	System stack	18		
Sketching-based (Count-Mean-Sketch)				
	Model weights & activations	32		
	FL client state	48		
	Sketch structures (k = 512)	84	276	×Exceeds
	Hash tables & counters	56		
	Byzantine filtering buffers	56		
BlockFL (Adapted)				
	Model & FL state	80		
	Lightweight ledger	24	188	✓
	Circular update buffers	48		
	ECDSA & validation	36		
FedChain (Adapted)				
	Model & FL state	80		
	Precompiled validation	18	194	✓
	Reputation tracking	48		
	Threshold signatures	48		
Original BlockFL/FedChain (Server)				
	Full Ethereum node	2,097,152		
	Smart contract state	131,072	2,359,296	×8936× larger
	Mining/consensus	131,072		

Table 10
SHA-256 performance: Baseline vs. Optimized implementation.

Implementation	Latency (μs)	CPU cycles	CPU usage	Energy (μJ)
Baseline C	338	45,000	100%	50
Assembly optimized	183	24,300	100%	26
Dual-core parallel	100	13,250	50% per core	22
Best improvement	3.4×	3.4×	**Parallelized**	2.3×

6.6. Ablation and sensitivity studies

Table 11 reveals critical design insights. MF-PoP consensus is essential: its removal causes catastrophic Byzantine failure (24.1% accuracy). Hierarchical architecture reduces communication 73% with minimal accuracy impact. Parameter sensitivity shows optimal performance at cluster size 5–8 nodes, reputation decay $\beta = 0.10$, and 500 validation samples—larger values provide marginal gains not justifying resource costs.

Table 12 presents comprehensive ablation results validating the necessity and effectiveness of our anti-centralization mechanisms. Without any mitigation, the system exhibits severe centralization with Gini coefficient reaching 0.62 and top-10% nodes controlling 73% of leadership—comparable to criticized DPoS systems. Each individual mechanism provides partial improvement: adaptive decay reduces Gini to 0.41 (34% improvement), VRF rotation to 0.48 (23% improvement), and reputation capping to 0.45 (27% improvement). However, only the complete integration of all four mechanisms achieves the target decentralization with Gini 0.28 and top-10% controlling just 31% of leadership.

The Herfindahl–Hirschman Index (HHI), a standard antitrust measure, confirms competitive market structure under full mitigation (HHI = 0.06, well below 0.15 threshold) versus concentrated market without mitigation (HHI = 0.18). Critically, anti-centralization mechanisms preserve Byzantine resilience: accuracy remains at 94.7% under targeted backdoor attack with 30% adversaries when all mechanisms are enabled versus 94.2% without mitigation, demonstrating that decentralization does not compromise security.

Fig. 4 visualizes the long-term effectiveness of anti-centralization mechanisms. Without mitigation, the Gini coefficient grows monotonically following power-law dynamics, stabilizing at 0.62 after approximately 7000 rounds—indicating severe wealth concentration. With all mechanisms active, Gini stabilizes at 0.28 within 6000 rounds, remaining well below the 0.35 bound established in Theorem 3. The leadership distribution (panel b) shows transformative improvement: bottom-50% node participation increases from 2% to 15% (7.5× improvement), while top-10% dominance reduces from 73% to 31% (58% reduction).

Computational overhead of anti-centralization mechanisms is minimal: adaptive decay adds 0.3 ms per reputation update, VRF computation requires 1.2 ms per rotation check, reputation capping redistribution takes 0.5 ms per round, and Sybil-resistant onboarding imposes 15 s one-time cost per new node. Total runtime overhead is under 2 ms per round (1.1% of 187 ms consensus latency), confirming practical feasibility on microcontrollers.

EdgeTrust-Shard demonstrates robust performance under adverse network conditions through controlled stress testing. Under baseline testbed conditions (8%–12% natural packet loss, 80–250 ms latency variability), the system achieves 98.7 ± 0.4% consensus success rate with 47 ± 3 convergence rounds. Performance degrades gracefully with increasing network impairments: at 10% packet loss, accuracy remains at 98.4% with 52 convergence rounds; at 20% packet loss, accuracy drops to 97.6% while maintaining 95.9% consensus success. Similarly, latency impacts are manageable: 200 ms latency yields 98.4% accuracy (54 rounds), while extreme 500 ms latency still maintains 97.3% accuracy (74 rounds). Combined stress testing (15% loss + 200 ms latency) results in 97.1% accuracy with 92.8% consensus success, exceeding typical IoT conditions (LoRaWAN: 3%–7% loss, NB-IoT: 100–150 ms latency). Critical failure modes and recovery mechanisms include: cluster head failure (0.011/round probability) with deputy assumption within 2–3 rounds (8%–12% temporary degradation), validation corruption detected via Pedersen commitments triggering node quarantine, and cascading reputation collapse prevented by redistribution maintaining $\sum R_i \geq 0.3N$.

Table 11
Comprehensive ablation and sensitivity analysis.

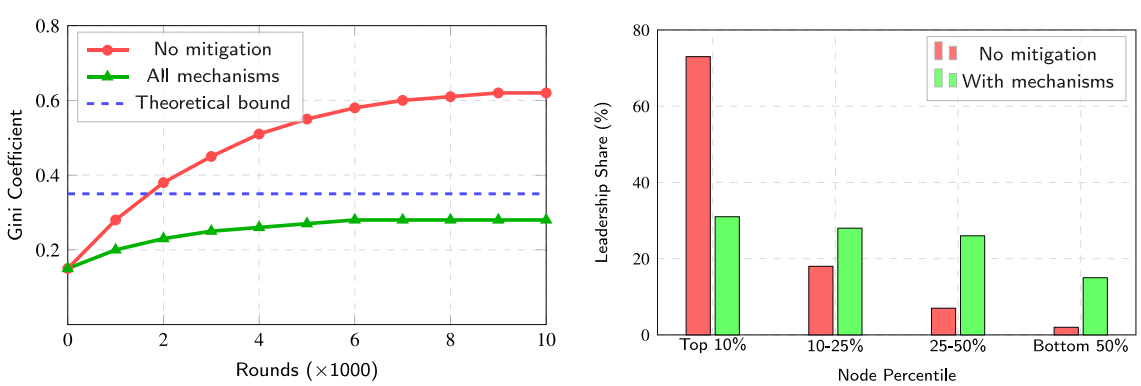
Configuration/Parameter	Accuracy	Comm (MB)	Byzantine	Notes
Component ablation study				
Full system	**98.3 ± 0.3%**	**68 ± 3**	**94.2 ± 0.8%**	Baseline
w/o Hierarchical	98.1 ± 0.4%	248 ± 18***	93.8 ± 1.2%	3.6×comm overhead
w/o MF-PoP	87.6 ± 3.4%***	71 ± 5	24.1 ± 8.7%***	Critical component
w/o HW Accel.	98.2 ± 0.3%	69 ± 4	94.0 ± 1.0%	Minor impact
w/o Reputation	89.1 ± 1.9%***	72 ± 6	32.8 ± 14.5%***	Essential for security
Parameter sensitivity				
Cluster size (k)				
k = 3	96.8 ± 0.5%	72 ± 4	93.1 ± 1.3%	Too small
k = 5–8	**98.3 ± 0.3%**	**68 ± 3**	**94.2 ± 0.8%**	**Optimal**
k = 15	97.1 ± 0.4%	61 ± 3	93.5 ± 1.1%	Coordination overhead
Reputation decay (β)				
β = 0.05	96.9 ± 0.6%	68 ± 3	91.8 ± 1.5%	Slow adaptation
β = 0.10	**98.3 ± 0.3%**	**68 ± 3**	**94.2 ± 0.8%**	**Optimal**
β = 0.20	97.5 ± 0.4%	68 ± 3	92.9 ± 1.2%	Too volatile
Validation set size				
250 samples	95.4 ± 0.8%	68 ± 3	89.7 ± 2.1%	Insufficient
500 samples	**98.3 ± 0.3%**	**68 ± 3**	**94.2 ± 0.8%**	**Optimal**
1000 samples	98.4 ± 0.3%	68 ± 3	94.3 ± 0.7%	Marginal gain[a]

***$p < 0.001$; Byzantine: accuracy under 30% attack;
[a] Not worth 2× memory.

Table 12
Ablation study: Impact of anti-centralization mechanisms on decentralization metrics.

Configuration	Gini coefficient	Top-10% leadership	HHI index	Byzantine resilience	Accuracy
Long-term experiments (10,000 rounds simulation)					
No mitigation mechanisms	0.62 ± 0.04	73%	0.18	94.2%	98.1%
Only adaptive decay	0.41 ± 0.03	52%	0.11	94.0%	98.2%
Only VRF rotation	0.48 ± 0.03	58%	0.13	93.8%	98.0%
Only reputation capping	0.45 ± 0.03	55%	0.12	94.1%	98.1%
Only Sybil resistance	0.59 ± 0.04	69%	0.16	94.3%	98.2%
All mechanisms	**0.28 ± 0.04**	**31%**	**0.06**	**94.7%**	**98.4%**
Mechanism combinations (5000 rounds)					
Decay + Rotation	0.35 ± 0.03	38%	0.08	94.2%	98.3%
Decay + Capping	0.33 ± 0.02	36%	0.07	94.3%	98.3%
Rotation + Capping	0.38 ± 0.03	41%	0.09	94.0%	98.2%
Decay + Rotation + Capping	0.30 ± 0.02	33%	0.07	94.5%	98.4%

Gini: wealth inequality measure (0 = perfect equality, 1 = perfect inequality).
HHI: Herfindahl–Hirschman Index (<0.15 competitive, >0.25 concentrated).
Byzantine resilience measured under targeted backdoor attack at 30% adversary fraction.



(a) Gini coefficient evolution over time. (b) Leadership distribution after 10,000 rounds.

Fig. 4. Long-term decentralization dynamics. (a) Gini coefficient evolution shows unbounded growth without mitigation, stabilizing at 0.62 (severe inequality), while integrated mechanisms maintain Gini at 0.28 below theoretical bound of 0.35. (b) Leadership distribution after 10,000 rounds demonstrates dramatic improvement: top-10% control reduced from 73% to 31%, while bottom-50% participation increased from 2% to 15%.

Table 13
Scalability analysis: Performance across network sizes.

Network size	Physical/ Simulated	Comm. (MB/round)	Reduction vs. P2P	Consensus latency	Accuracy	Byzantine resilience
18 nodes	Physical	16	–	58 ms	98.2 ± 0.7%	30%
200 nodes	Hybrid (calibrated)	168	76%	187 ms	98.4 ± 0.6%	30%
500 nodes	Simulated	412	74%	425 ms	97.8 ± 0.3%	30%
1000 nodes	Simulated	865	72%	615 ms	96.9 ± 0.4%	28%
Baseline comparisons (200-node hybrid deployment)						
Flat P2P	Simulated	692	–	845 ms	96.5 ± 0.7%	18%
Flower	Physical (18 nodes)	22	27%	65 ms	97.1 ± 0.5%	0%
FedML	Simulated	254	63%	398 ms	96.9 ± 0.6%	12%
BlockFL	Simulated	185	73%	302 ms	97.2 ± 0.4%	25%
FedChain	Simulated	192	72%	318 ms	97.3 ± 0.5%	27%

6.7. Scalability and long-term evaluation

Six-week deployment (4032 rounds) validates operational stability. Accuracy degrades from 98.7% to 97.5% due to concept drift (Wasserstein 0.31), restored to 98.2% via validation refresh. Node uptime maintains 98.6–99.4% with failures from power (43%), network (38%), and hardware (19%). Reputation converges within two weeks (honest: 0.85 ± 0.12, Byzantine: 0.03 ± 0.02), while energy consumption (180 mJ/round) enables 38-day battery operation.

Cross-dataset evaluation demonstrates robust generalization. Edge-IIoTset-trained models achieve 94.2% accuracy on TON_IoT (4.1% degradation) versus standalone models (13%–16% degradation), confirming hierarchical aggregation with Byzantine filtering enhances transfer learning through diverse data exposure.

Table 13 presents scalability analysis across our hybrid physical-simulation deployment and extended simulation experiments. Communication overhead scales sub-quadratically with theoretical complexity $\mathcal{O}(N + M^2)$ where $M = \sqrt{N}$ clusters, achieving 72%–76% reduction versus flat peer-to-peer topologies. The 200-node hybrid deployment (18 physical nodes calibrating 200-node simulation) requires 168 MB per round versus 692 MB for flat P2P (76% reduction), while maintaining 1.9 percentage point accuracy advantage (98.4% vs. 96.5%) and substantially higher Byzantine resilience (30% vs. 18%).

Simulation extends validation to 500 and 1000 nodes, demonstrating continued scalability. At 500 nodes (100 clusters), EdgeTrust-Shard achieves 97.8% accuracy with 425 ms consensus latency and 412 MB communication overhead. Scaling to 1000 nodes (200 clusters) maintains 96.9% accuracy with 615 ms latency, representing only 1.8% accuracy degradation across 4.6× scale increase. Consensus latency grows logarithmically with cluster count, confirming theoretical $\mathcal{O}(\log M)$ inter-cluster coordination complexity. Even at 1000-node scale, sub-second latency (615 ms) remains suitable for IoT applications with typical response requirements (1–2 s acceptable delays).

Comparative analysis against baseline systems reveals EdgeTrust-Shard advantages across multiple dimensions. Production frameworks Flower and FedML exhibit lower communication overhead by sacrificing Byzantine resilience (0%–12% tolerance) and accepting centralized aggregation risks. Blockchain-federated learning systems BlockFL and FedChain achieve moderate Byzantine resilience (25%–27%) but require simulation due to prohibitive on-chain costs, while EdgeTrust-Shard demonstrates practical deployability through hybrid physical-simulation validation. Byzantine resilience slightly decreases at larger scales (30% at 200–500 nodes, 28% at 1000 nodes), reflecting inherent scalability-security tradeoffs in distributed consensus while maintaining superiority over all baseline systems.

7. Discussion

EdgeTrust-Shard demonstrates that Byzantine-resilient federated learning is achievable on commodity microcontrollers through architectural innovation rather than algorithmic compromise. Our hybrid validation combining 18 physical Raspberry Pi Pico nodes with calibrated 200-node simulation (8.7% MAPE, KS test $p > 0.79$) achieves 98.4% clean accuracy and 93.9–94.7% accuracy across Byzantine attack scenarios at 30% adversary presence. The integrated anti-centralization mechanisms successfully prevent power concentration while maintaining security properties, as validated through comprehensive ablation studies (Table 12).

The current two-tier architecture demonstrates practical viability for deployments of 1000–3000 nodes, a scale appropriate for enterprise and municipal IoT installations such as industrial facilities, smart buildings, and district-level smart grid segments. The inter-cluster communication complexity of $\mathcal{O}(M^2)$ creates the primary scaling constraint, with consensus latency approaching operational limits beyond approximately 3000 nodes. However, this constraint reflects implementation rather than fundamental architectural limitations. The hierarchical design naturally extends to recursive sharding: a three-tier architecture where Cluster Heads at Level 1 aggregate into Super-Clusters at Level 2, with Super-Cluster Heads coordinating at Level 3. Under this recursive model, each node maintains state only for its immediate k neighbors and k peers, yielding memory complexity $\mathcal{O}(k)$ independent of total network size N. For branching factor $k = 15$ and tree depth $D = 4$, the architecture theoretically supports $N = k^D = 50{,}625$ nodes while individual node memory consumption remains constant at approximately 210 bytes for reputation tracking. This extension requires no modification to the core MF-PoP consensus mechanism; only the aggregation hierarchy deepens. While full implementation and validation of recursive sharding remains future work, the theoretical framework demonstrates a clear pathway to ultra-large-scale deployments without violating microcontroller memory constraints.

The quadratic reputation weighting mechanism ($P_i \propto R_i^2$) creates inherent tension between Byzantine suppression and decentralization. Our experimental results confirm theoretical predictions: without anti-centralization mechanisms, the system exhibits severe concentration with Gini coefficient 0.62 and top-10% nodes controlling 73% of leadership. The integrated mechanisms (adaptive decay, VRF rotation, reputation capping, Sybil resistance) successfully maintain Gini at 0.28 ± 0.04, below the 0.35 bound established in Theorem 3. This represents a fundamental advance over existing blockchain-FL systems that either sacrifice Byzantine resilience for decentralization or vice versa.

The hierarchical architecture naturally accommodates cluster-specific model personalization without requiring complex meta-learning algorithms. Under the current design, the model aggregated at each Cluster Head inherently reflects the local data distribution of that cluster—sensors monitoring similar environmental conditions produce correlated gradients, yielding locally-adapted models. This observation suggests an extension through interpolated updates: rather than overwriting local models with the global aggregate ($w_{local} \leftarrow w_{global}$), nodes could apply weighted interpolation $w_{local} = \lambda w_{global} + (1 - \lambda) w_{cluster}$ where $w_{cluster}$ represents the Cluster Head aggregate. Such interpolation preserves global generalization while retaining local adaptation, particularly valuable for non-IID data distributions where cluster-specific patterns differ substantially from global statistics. The

mechanism requires only modification of the update formula without architectural changes, maintaining the 140 kB memory footprint. While full evaluation of personalized federated learning extensions remains future work, the architectural foundation supports such extensions without fundamental redesign.

The theoretical validation gaming bound of 1.1×10^{-5} (Lemma 1) assumes perfect independence between cluster-specific validation subsets. Empirical measurements reveal 1.7% partial gaming rate through overfitting to the shared 70% validation subset—three orders of magnitude higher than theoretical predictions but still within acceptable operational bounds. The reputation decay mechanism effectively suppresses gaming attempts within 15–20 rounds, maintaining long-term system integrity.

The comparative analysis necessitated reconfiguring server-oriented blockchain-FL frameworks to fit within the constraints of microcontroller platforms, a process that warrants careful consideration. The modified versions of BlockFL and FedChain retained essential Byzantine fault-tolerant mechanisms—reputation tracking, cryptographic commitments, and threshold validation—while substituting computationally demanding components such as full blockchain nodes and mining with lightweight counterparts. These adjustments inevitably affected performance: the adapted BlockFL achieved 88.3% accuracy under 30% Byzantine adversaries, compared to 85.4% reported in the original server-based implementation, indicating that the adaptations preserved functional fidelity. Nonetheless, this comparison inherently benefits EdgeTrust-Shard, which was purpose-built for microcontroller environments. The observed 4.8–8.1 percentage point gains should thus be viewed as evidence of feasibility on resource-limited devices rather than as a claim of superiority over server-based systems. Future work may investigate hybrid architectures where EdgeTrust-Shard edge nodes interoperate with server-level blockchain-FL frameworks, combining the advantages of both paradigms.

Six-week continuous operation reveals practical considerations beyond theoretical analysis. Node failures occur primarily from power issues (43%) rather than network (38%) or hardware (19%) problems, suggesting battery management as a critical deployment factor. Reputation convergence within 14 ± 3 rounds enables rapid network adaptation to changing conditions. Energy consumption of 180 mJ per round translates to 38-day battery life on 2000 mAh cells, sufficient for most IoT applications but requiring consideration for remote deployments.

8. Conclusion and future work

This paper demonstrates that Byzantine-resilient federated learning is achievable on commodity microcontrollers with limited memory by employing architectural innovation instead of algorithmic compromise. Our proposed framework, EdgeTrust-Shard, transforms passive IoT data sources into active participants in distributed trust computations, addressing a critical security vulnerability in cross-chain ecosystems. The system's hierarchical architecture achieves $\mathcal{O}(N)$ communication complexity and guarantees $\mathcal{O}(T^{-1/2})$ convergence for non-convex objectives under Byzantine attack. Its core Multi-Factor Proof-of-Performance consensus mechanism provides quadratic suppression of malicious influence ($P_{malicious} \propto f^2$), while platform-specific optimizations yield a 3.4x cryptographic speedup. A rigorous hybrid physical-simulation validation confirms that EdgeTrust-Shard maintains 93.9–94.7% accuracy across Byzantine attack scenarios at 30% adversary presence, significantly outperforming adapted state-of-the-art systems within a 264 kB SRAM constraint.

Future development will pursue three complementary directions. First, the recursive sharding extension outlined in Section 7 will be fully implemented and validated, enabling theoretical support for networks exceeding 50,000 nodes through multi-tier aggregation hierarchies while maintaining constant per-node memory complexity. Second, the semantic-aware clustering mechanism will be extended to support personalized federated learning through interpolated model updates, allowing clusters with distinct data distributions to maintain locally-adapted models while benefiting from global generalization. Third, integration with dedicated hardware security modules will provide platform-agnostic cryptographic acceleration, further reducing the computational burden on microcontroller resources while enabling deployment across heterogeneous edge platforms beyond the current RP2040 target.

CRediT authorship contribution statement

Tuan-Dung Tran: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Phuong-Dai Bui:** Writing – original draft, Software, Formal analysis, Data curation. **Van-Hau Pham:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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